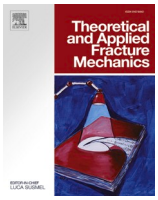
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Abaqus implementation of coupled thermo-mechanical dual-horizon bond-based peridynamics for thermally induced fractures in thermal barrier coatings

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ABSTRACT

Thermal barrier coatings (TBCs) are responsible for thermal insulation protection of engine turbine blades in the industrial gas turbines and aero-engines. However, it often suffers from the thermal shocks during the normal service, which may cause the performance loss and delamination failure of TBCs. Thus, it is significant to study the thermally induced fracture mechanisms of TBCs. To this end, we propose a coupled thermo-mechanical dual-horizon bond-based peridynamics (TM-DH-BPD) with the optimum mathematical expression of micro-thermal conductivity. Meanwhile, we develop the dual-horizon peridynamic finite element method (DHPD-FEM) along with two approaches of the implicit solver and explicit solver with mass scaling to effectively address the time scale gap of the coupled thermo-mechanical governing equations. Moreover, the two and three-dimensional TM-DH-BPD models have been firstly implemented in Abaqus UEL and VUEL subroutines. At last, the TM-DH-BPD model is used to investigate the thermally induced fracture mechanism of TBCs, the phenomena of the transverse cracking propagation, kinking in the top coat and its interaction with the TC/BC (Top coat/Bond coat) interfacial cracking observed in the experiment are well captured. The proposed TM-DH-BPD model and its dual-horizon finite element implementation in the commercial software may accelerate the pace of the theoretical researching and real-word application of peridynamics including the complex thermal shock cracking problems of TBCs.

1. Introduction

Thermal barrier coatings (TBCs), commonly used in the industrial gas turbines and aero-engines because of their excellent thermodynamic properties of high temperature and corrosion resistances, are responsible for thermal insulation protection of engine turbine blades [1–3]. During the normal service, they are subjected to the high temperature and thermal shocks from gases in the turbine engine. It will lead to the performance loss and delamination failure of thermal barrier coatings in these cases. Therefore, to better evaluate its service performance from the mechanical point of view, and accurately understand and reveal the interface damage mechanism under thermal shocks, it is of great significance to develop the multi-physics coupled damage model, calculational method and CAE software.

Important works have been done to study the damage mechanisms of TBCs under thermal shocks. Liu et al. [4] combined the thermal shock test and finite element model (FEM) to study the main damage modes and its life prediction of TBCs. The results show that the spalling failure is comprehensively influenced by the interface debonding and surface damage of TBCs. Periodic surface damages and interface debonding of TBCs can be observed in the experiment of the cooling process [5], Fan et al. [5] used FEM along with cohesive zone model (CZM) to investigate

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the influence of the periodic surface damages on the interface debonding of TBCs. They found that the distribution of the short interface damage was greatly influenced by the surface crack spacing. Recently, Liang et al. [3] carried out the thermal shock cycling tests to study the influence of the top coat thickness on the spallation failure of TBCs. Meanwhile, they investigated the coating surface stress distribution by using two-dimensional FEM. The simulation results indicated that the tensile stress in the coating surface dominates during the thermal shock of heating. And the compressive stress state dominates during the thermal shock of quenching. Current numerical approaches have been proposed to address the relatively simple damage problems of thermal barrier coatings [6]. However, the phenomena of the transverse cracking propagation, kinking and its interaction with the TC/BC interfacial cracking observed in the experiment of the cooling process [3,5,7] has not been well captured. To address the thermally induced interface damage initiation and propagation in TBCs, Wang et al. [6] presented a peridynamics-based coupled thermo-mechanical damage model to investigate the cracking behaviors in TC layer. They stated that the peridynamic models are potential for studying the cracking behavior in TBCs. Wu et al. [8] discussed the impact of the thermal shock time and interface morphology on the stress distribution of TBCs and the interface debonding. They also stated that the peridynamic model may be an effective tool for investigating the interface debonding of TBC. Although some attempts have been done to establish the peridynamic models to address the thermally induced fractures during thermal shocks in TBCs, two-dimensional peridynamic models are commonly established that it cannot account for the significant effect of the thermal shocks from the front and back sides of TBCs on the fracture behaviors. Additionally, the transient temperature diffusion cannot be described for the above-mentioned two peridynamic models. Thus, they cannot capture the multiple transverse cracks that initiate on the coating surface and propagate toward the TC/BC interface observed in the experiment of thermal shocks on the coating surface [3,5,7]. To comprehensively explain the complex thermally induced fracture mechanism, it is important to propose the effective peridynamics-based coupled thermo-mechanical damage model that considers the heat conduction and interface damage criterion.

As the nonlocal theory, Peridynamics (PD) has obtained great interest in the field of damage mechanics [9]. Compared with the classical theory based on the partial differential equation, the peridynamic model is on the basis of the spatial integral equation. Since the peridynamic spatial integral governing equation is valid at damage surfaces, the damage can be easily modeled without any complex treatments. As with the meshfree models presented by Rabczuk et al. [10–12], the peridynamics-based damage models has great advantages in dealing with the damage problems. Heretofore, PD has been extensively validated to effectively deal with various fracture problems for homogeneous and heterogeneous materials [13–22], the chemical-induced cracking problems [23–25], the cross-scale cracking problems [26,27], the nonlinear material problems [28–34] and finite deformation problems [35–38]. Additionally, some notable thermo-mechanical damage models in the context of peridynamics were proposed to investigate the thermally-induced deformation and cracking problems. Oterkus et al. [39] established the explicit peridynamics-based coupled thermo-mechanical model on the basis of thermodynamic considerations. He et al. [40] proposed modified peridynamics-based thermo-mechanical damage model to investigate the deformation and thermally-induced failure of functionally graded materials. Liu et al. [41] proposed the thermo-mechanical peridynamic damage model to study the damage processes of nuclear fuel pellets. Bazazzadeh et al. [42] established the adaptive peridynamics to study the damage propagation of ceramics. In the peridynamics-based thermo-mechanical damage models with the explicit solver, the multiple time steps are used to address the time scale gap. However, the minimum time step is limited when using the multiple time steps with the explicit solver. To address the steady-state damage propagation problems in the context of peridynamics, Adaptive Dynamic Relaxation (ADR) [43,44] is generally employed. As for addressing the interface damage and debonding of TBCs, dual-horizon peridynamics (DH-PD) will be employed in this paper. Dual-horizon peridynamics, which is suitable to use variable peridynamic horizon and the nonuniform discretization [45,46], has been presented by Ren et al. [47,48]. Thanks to the damage criterion that the peridynamic bond and dual-bond can be damaged separately in the context of dual-horizon peridynamics, DH-PD is able to naturally address the interface failure and debonding problems without the extra treatments [47,49,50]. Bie et al. [51] have proposed dual-horizon peridynamics-based correspondence principle on the basis of DH-PD and applied it to deal with the interface debonding for the heterogeneous materials [52,53]. To take the advantage of DH-PD in characterizing the interface debonding of the heterogeneous materials, the dual-horizon peridynamics-based coupled thermo-mechanical damage model will be established to address thermally induced fractures including the complex transverse cracking propagation, kinking and interface debonding in thermal barrier coatings.

Since peridynamics is the nonlocal model, the computational efficiency is big challenging compared with FEM. Coupling peridynamics and FEM may be meaningful to take full advantages of PD to address the damage propagation problems and the calculation efficiency of FEM [54–60]. Some other coupling models of isogeometric-meshfree method and peridynamics can be seen in the references [61, 62]. Another potentially valuable and meaningful method for researchers and engineers may be the integration of the peridynamic models in the commercial software, e.g. Abaqus and Ansys. It can use the powerful computing power and post-processing visualization capabilities of commercial software. Recently, some studies have integrated the peridynamic models into the commercial software to enhance the computational efficiency for the real-world engineering applications [63]. In 2007, Macek and Silling [64] implemented bond-based peridynamics in Abaqus by using truss element. Afterwards, in 2020, Bie et al. [65] firstly proposed the concept of the peridynamic node and element, and then implemented dual-horizon peridynamics and its coupled model between DH-PD and FEM in Abaqus UEL/UMAT subroutines of Abaqus/Standard and VUEL/VUMAT subroutines of Abaqus/Explicit. Additionally, they proposed the identical mapping relation to display the calculated results of the damage distribution in the visualization module of Abaqus [65]. In which 2D and 3D numerical examples including contact and damage are considered. In 2022, Anicode and Madenci [66] implemented peridynamics in Ansys with the native MATRIX27 elements; Han et al. [67] implemented the peridynamics-based FEM in Abaqus and Ansys for the brittle fractures. In 2024, Anicode and Madenci [68] implemented the coupling model of DH-PD and FEM for the irregular discretization in Ansys. In 2025, Shen et al. [69] implemented the peridynamics-based FEM of thin shells to address brittle fractures in Abaqus, they use the identical mapping relation [65] to visualize the damage plots in Abaqus. Note that the above peridynamic models in Abaqus or Ansys are applied to address the brittle fracture problems of the homogeneous materials under mechanical loads. It is significantly meaningful to implement the peridynamics-based coupled thermo-mechanical damage models in the commercial software to deal with thermally-induced damage problems. Which includes the complex transverse cracking propagation, kinking and interface debonding in thermal barrier coatings.

Firstly, we will develop the coupled thermo-mechanical dual-horizon bond-based peridynamics (TM-DH-BPD) with the optimum mathematical expression of micro-thermal conductivity. Then, the discretized forms of the linearized integral heat conduction equation and the linearized integral equilibrium equation of motion are established. To deal with the time scale gap and numerical accuracy of TM-DH-BPD, the authors propose the dual-horizon peridynamic finite element method (DHPD-FEM) along with two approaches of the standard implicit solver in Abaqus/standard and explicit solver with mass scaling in Abaqus/Explicit. In this way, the relatively large time step can be used. Moreover, the dual-horizon peridynamic finite element method is

implemented in Abaqus/UEL and Abaqus/VUEL to address the coupled thermo-mechanical dual-horizon bond-based peridynamic equation. Compared with Adaptive Dynamic Relaxation (ADR), the proposed DHPD-FEM with the standard implicit solver may be more accurate to solve for the coupled peridynamics-based thermo-mechanical equations. Additionally, since it is convenient to implement the proposed DHPD-FEM in Abaqus, it can use the powerful parallel computing power and visualization capabilities of Abaqus. Finally, the usefulness of the TM-DH-BPD model and its corresponding implementation of dual-horizon peridynamic finite element method (DHPD-FEM) in Abaqus are validated with some damage examples. Which includes the thermal shock induced failure of ceramic specimen and the thermally induced interface damage and debonding of TBCs. The phenomena of the transverse cracking propagation, kinking in the top coat and its interaction with the TC/BC interfacial cracking that was observed in the thermal shock experiment are well captured. Noted that the TM-DH-BPD model has been firstly implemented in Abaqus using DHPD-FEM. With the same method, other similar multi-physical coupled model (e.g., mechano-chemical, thermo-mechano-chemical and thermo-electrical et al.) can be solved by the proposed the dual-horizon peridynamic finite element method and implemented in Abaqus to address the corresponding complex multi-physical coupled engineering problems with the interface debonding and damage propagation in the context of peridynamics. It may accelerate the pace of the theoretical researching and real-word application of peridynamics including the complex thermal shock cracking problems of TBCs.

2. The TM-DH-BPD model

2.1. The dual-horizon bond-based peridynamics for mechanical interaction

In this section, a brief outline of dual-horizon bond-based peridynamics is provided. More detail information can be found in the references [47, 48]. The initial configuration of the computational domain $\Omega_0 \subset R^n (n = 1, 2, 3)$ is shown in Fig. 1. The peridynamic horizon length of a material point $\mathbf{x} \in \Omega_0$ is denoted by $\delta_x > 0$. For another peridynamic material point $\mathbf{x}' \in \Omega_0$, if $|\mathbf{x}' - \mathbf{x}| \le \delta_x$, then the peridynamic material point $\mathbf{x}'$ belongs to the family of the peridynamic material point $\mathbf{x}$, such that $\mathbf{x}' \in \mathscr{H}_x$. The peridynamic horizon is defined in the form of

$$\mathscr{H}_x = \{\mathbf{x}' \in \Omega_0, |\mathbf{x}' - \mathbf{x}| \le \delta_x\} \tag{1}$$

Correspondently, the relative displacement vector $\boldsymbol{\eta}$ is defined by

$$\boldsymbol{\eta} = \underline{\mathbf{U}}\langle\boldsymbol{\xi}\rangle = \mathbf{u}(\mathbf{x}', t) - \mathbf{u}(\mathbf{x}, t) \tag{2}$$

where $\mathbf{u}(\mathbf{x}', t)$ and $\mathbf{u}(\mathbf{x}, t)$ denote the displacement vectors associated with the material points $\mathbf{x}'$ and $\mathbf{x}$, respectively. $\underline{\mathbf{U}}\langle\boldsymbol{\xi}\rangle$denotes the relative displacement state with one underline. The peridynamic state depending on position and time is the mathematical object used to map vectors into general quantities such as a scalar, a vector or a tensor. It can be seen that the relative displacement "state" has the same property of the relative displacement "vector".

The relative position vector in the current configuration is denoted by

$$\mathbf{y}(\mathbf{x}', t) - \mathbf{y}(\mathbf{x}, t) = \boldsymbol{\xi} + \boldsymbol{\eta} \tag{3}$$

where $\mathbf{y}(\mathbf{x}, t)$ and $\mathbf{y}(\mathbf{x}', t)$ denote the spatial coordinate vectors associated with the material points $\mathbf{x}$ and $\mathbf{x}'$ in the current configuration.

The integral equilibrium equation of motion for a material point $\mathbf{x}$ in the context of dual-horizon bond-based peridynamics [47,48] can be given by

$$\rho\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_{\mathscr{H}_x} \underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle \mathrm{d}V_{x'} - \int_{\mathscr{H}'_x} \underline{\mathbf{T}}[\mathbf{x}', t]\langle -\boldsymbol{\xi}\rangle \mathrm{d}V_{x'} + \mathbf{b}(\mathbf{x}, t) \tag{4}$$

where ρ denotes the density and $V_{x'}$ denotes the volume of material point $\mathbf{x}'$. $\ddot{\mathbf{u}}$ denotes the acceleration vector. $\mathbf{b}$ denotes the body force. $\underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle$ denotes the peridynamics-based direct force state that is applied on material point $\mathbf{x}$ in the horizon $\mathscr{H}_x$. Correspondently, $\underline{\mathbf{T}}[\mathbf{x}', t]\langle -\boldsymbol{\xi}\rangle$ denotes the peridynamics-based reaction force state that is applied on material point $\mathbf{x}$ in the dual-horizon $\mathscr{H}'_x$. The dual-horizon of a material point $\mathbf{x}$ is denoted by

$$\mathscr{H}'_x = \{\mathbf{x}' \in \Omega_0 | \mathbf{x} \in \mathscr{H}_{x'}\} \tag{5}$$

For bond-based peridynamics [47,48], the force state $\underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle$ can be written as

$$\underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle = \frac{1}{2} c \cdot s\langle\boldsymbol{\xi}\rangle \cdot \frac{\boldsymbol{\xi} + \boldsymbol{\eta}}{|\boldsymbol{\xi} + \boldsymbol{\eta}|} \tag{6}$$

For dual-horizon bond-based peridynamics, $\underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle = -\underline{\mathbf{T}}[\mathbf{x}', t]\langle -\boldsymbol{\xi}\rangle$. Therefore, the integral equilibrium equation of motion in Eq. (4) can be changed as

$$\rho\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_{\mathscr{H}_x} \underline{\mathbf{T}}[\mathbf{x}, t]\langle\boldsymbol{\xi}\rangle \mathrm{d}V_{x'} + \int_{\mathscr{H}'_x} \underline{\mathbf{T}}[\mathbf{x}', t]\langle\boldsymbol{\xi}\rangle \mathrm{d}V_{x'} + \mathbf{b}(\mathbf{x}, t) \tag{7}$$

where the scalar bond strain $s\langle\boldsymbol{\xi}\rangle$ is written as

$$s\langle\boldsymbol{\xi}\rangle = (|\boldsymbol{\xi} + \boldsymbol{\eta}| - |\boldsymbol{\xi}|)/|\boldsymbol{\xi}| \tag{8}$$

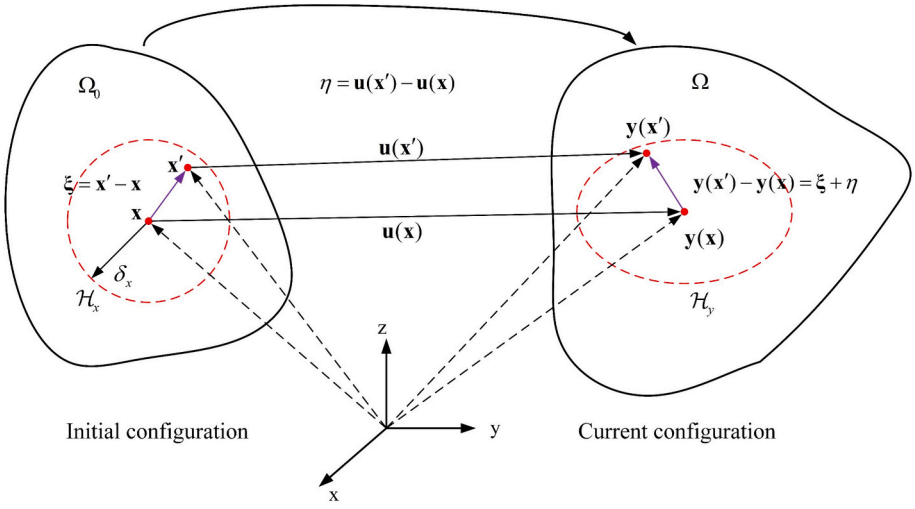


Fig. 1. Schematic diagram of material point movement for dual-horizon bond-based peridynamics.

The micro-modulus c is also called the stiffness of the bond. The final mathematical expression [56] is given by

$$c=\begin{cases}\dfrac{12E}{\pi\delta_x^4} & \text{for 3D}\\ \dfrac{9E}{\pi H\delta_x^3} & \text{for plane stress}\\ \dfrac{48E}{5\pi H\delta_x^3} & \text{for plane strain}\end{cases} \tag{9}$$

where E denotes elastic modulus. H denotes thickness of the two-dimensional problems.

Take the initial configuration as the reference configuration, the discretized form of the linearized integral equilibrium equation of motion in Eq. (7) is expressed as

$$\rho\ddot{\mathbf{u}}(\mathbf{x},t)=\left(\sum_{\mathbf{x}'\in\mathcal{H}_x}\underline{\underline{\mathbf{K}}}^{u}[\mathbf{x}]+\sum_{\mathbf{x}'\in\mathcal{H}'_x}\underline{\underline{\mathbf{K}}}^{u}[\mathbf{x}']\right)\cdot\boldsymbol{\eta}\cdot\mathrm{V}_{x'}+\mathbf{b}(\mathbf{x},t) \tag{10}$$

where $\underline{\underline{\mathbf{K}}}^{u}[\mathbf{x}]$ can be seen as the mechanical stiffness (or called the double state with two underline) of the peridynamic bond $\boldsymbol{\xi}$ associated with the material point $\mathbf{x}$:

$$\underline{\underline{\mathbf{K}}}^{u}[\mathbf{x}]=\frac{\partial\underline{\mathbf{T}}[\mathbf{x},t=0]\langle\boldsymbol{\xi}\rangle}{\partial\boldsymbol{\eta}}\Big|_{\boldsymbol{\eta}=\mathbf{0}}=\frac{\partial[\mathbf{x},t=0]\langle\boldsymbol{\xi}\rangle}{\partial\underline{\mathbf{U}}\langle\boldsymbol{\xi}\rangle}\Big|_{\underline{\mathbf{U}}\langle\boldsymbol{\xi}\rangle=\mathbf{0}}=\frac{c}{2|\boldsymbol{\xi}|}\frac{\boldsymbol{\xi}}{|\boldsymbol{\xi}|}\otimes\frac{\boldsymbol{\xi}}{|\boldsymbol{\xi}|} \tag{11}$$

2.2. The dual-horizon bond-based peridynamics for heat conduction

The local form of heat transfer Eq. [44] is given by

$$\rho\cdot C_v\cdot\dot{\theta}(\mathbf{x},t)=\kappa\cdot\Delta\theta+r_q(\mathbf{x},t) \tag{12}$$

where r_q is the rate of heat generation per unit volume. $\theta(\mathbf{x},t)$ denotes the temperature of material point $\mathbf{x}$ at time t. C_v denotes the heat capacity. κ denotes the heat conduction coefficient. Δ is Laplacian operator.

Similar to the integral equilibrium equation of motion in Eq. (7), the integral equilibrium governing equation for the thermal part on the basis of dual-horizon bond-based peridynamics (DH-BPD) can be written as

$$\rho\cdot C_v\cdot\dot{\theta}(\mathbf{x},t)=\int_{\mathcal{H}_x}\underline{h}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle\mathrm{dV}_{x'}+\int_{\mathcal{H}'_x}\underline{h}[\mathbf{x}',t]\langle\boldsymbol{\xi}\rangle\mathrm{dV}_{x'}+r_q(\mathbf{x},t) \tag{13}$$

where $\underline{h}[\mathbf{x},t]$ is defined as the peridynamic heat flow state that is conjugate to the temperature state $\underline{\theta}\langle\boldsymbol{\xi}\rangle$ (the difference of two material points, $\underline{\theta}\langle\boldsymbol{\xi}\rangle=\theta(\mathbf{x}')-\theta(\mathbf{x})$).

It can be seen from Eqs. (12) and (13) that the fundamental distinction in mathematical formulation is the part of the heat conduction. The PDE-based heat transfer equation in Eq. (12) is partial differential form, which requires adequate smoothness of the field variables. Since the dual-horizon peridynamics-based heat transfer equation in Eq. (13) is integral form, it involves no derivatives with respect to the spatial coordinates in its heat conduction equation such that no adequate smoothness of the field variables is required. Therefore, the peridynamics-based heat transfer equation still holds when the computational fields are discontinuous or the cracks exist. Additionally, the other fundamental distinction in physical interpretation is that the PDE-based heat conduction occurs only between the adjacent material points (or finite element), we just call it local heat conduction function. However, the heat conduction occurs within the horizon of material points, we call it non-local heat conduction function. The degree of non-local heat conduction function is determined by the peridynamic horizon size.

Thanks to the non-local nature of peridynamics-based heat conduction function, the peridynamics-based model in temperature field calculation across the entire computational domain can not only accurately predict temperature distribution, but also handle discontinuities (e.g., cracks) without additional boundary treatments. Additionally, the other advantage of the peridynamics-based heat conduction model is the adaptability to deal with the heterogeneous materials and naturally address interface cracks without extra treatments.

Similar to the mathematical expression form of $\underline{\mathbf{T}}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle$ for the mechanical part in Eq. (6), the peridynamic heat flow state $\underline{h}[\mathbf{x},t]$ is defined in the form of

$$\underline{h}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle=-\underline{h}[\mathbf{x},t]\langle-\boldsymbol{\xi}\rangle=\frac{1}{2}k(\mathbf{x},\mathbf{x}')\cdot\frac{\underline{\theta}\langle\boldsymbol{\xi}\rangle}{|\boldsymbol{\xi}|^{\beta}} \tag{14}$$

where $\underline{\theta}\langle\boldsymbol{\xi}\rangle$ is the temperature state defined as the temperature difference of two material points. β is the shape parameter of micro-thermal potential in Appendix A. $k(\mathbf{x},\mathbf{x}')$ denotes the micro-thermal conductivity that has the following form

$$k(\mathbf{x},\mathbf{x}')=\begin{cases}k_0 & \text{Constant}\\ k_0(1-(\xi/\delta_x)^{\gamma}) & \text{Nonlinearity}\end{cases} \tag{15}$$

where k_0 is constant. γ denotes the shape parameter of the micro-thermal conductivity. Its mathematical expression can be derived by equating the nonlocal peridynamics-based thermal potential and local thermal potential in Appendix A.

When the constant $k(\mathbf{x},\mathbf{x}')$ is used, its mathematical expression for the 2D cases is

$$k(x,x')=\begin{cases}4\kappa/\left(\pi\cdot H\cdot\delta_x^2\right);\beta=2\\ 6\kappa/\left(\pi\cdot H\cdot\delta_x^3\right);\beta=1\\ 7\kappa/\left(\pi\cdot H\cdot\delta_x^{3.5}\right);\beta=0.5\end{cases} \tag{16}$$

where κ denotes the thermal conductivity coefficient. Bobaru et al. [70,71] states that the shape parameter γ will influence the numerical accuracy of the peridynamics-based heat conduction model. When the nonlinear micro-thermal conductivity is assumed, $k(\mathbf{x},\mathbf{x}')=k_0(1-(\xi/\delta_x)^{\gamma})$, the micro-thermal conductivity for the 2D case of $\gamma=1$ is

$$k(x,x')=\begin{cases}12\kappa/\left(\pi\cdot H\cdot\delta_x^2\right)\cdot(1-\xi/\delta_x);\beta=2,\gamma=1\\ 24\kappa/\left(\pi\cdot H\cdot\delta_x^3\right)\cdot(1-\xi/\delta_x);\beta=1,\gamma=1\\ 63\kappa/\left(2\cdot\pi\cdot H\cdot\delta_x^{3.5}\right)\cdot(1-\xi/\delta_x);\beta=0.5,\gamma=1\end{cases} \tag{17}$$

Take $\gamma=0.5$, $\beta=1$ as an example, the micro-thermal conductivity $k(\mathbf{x},\mathbf{x}')$ for the 3D case can be obtained as

$$k(x,x')=54\kappa\Big/\left(\pi\cdot\delta_x^4\right)\cdot\left(1-(\xi/\delta_x)^{0.5}\right);\beta=1,\gamma=0.5 \tag{18}$$

The micro-thermal conductivity can be obtained for different shape parameters γ of the micro-conductivity function in Eq. (15) and different shape parameters β of micro-thermal potential in Eq. (14). The detailed derivations are provided in Appendix A. The shape parameters β and γ will influence the numerical accuracy of the peridynamics-based heat conduction model. The optimum parameters β and γ are discussed and recommended in the following parts.

The discretized form of the linearized integral heat conduction equation is given by

$$\rho\cdot C_v\cdot\dot{\theta}(\mathbf{x},t)=\left(\sum_{\mathbf{x}'\in\mathcal{H}_x}\underline{\underline{K}}^{\theta}[\mathbf{x}]+\sum_{\mathbf{x}'\in\mathcal{H}'_x}\underline{\underline{K}}^{\theta}[\mathbf{x}']\right)\cdot\underline{\theta}\langle\boldsymbol{\xi}\rangle\cdot\mathrm{V}_{x'}+r_q(\mathbf{x},t) \tag{19}$$

where $\underline{\underline{K}}^{\theta}[\mathbf{x}]$ can be seen as the thermal bond stiffness of the peridynamic bond $\langle\boldsymbol{\xi}\rangle$,

$$\underline{\underline{K}}^{\theta}[\mathbf{x}]=\frac{\partial\underline{h}[\mathbf{x},t=0]\langle\boldsymbol{\xi}\rangle}{\partial\underline{\theta}\langle\boldsymbol{\xi}\rangle}\Big|_{\underline{\theta}\langle\boldsymbol{\xi}\rangle=\mathbf{0}}=\frac{k(\mathbf{x},\mathbf{x}')}{2|\boldsymbol{\xi}|^{\beta}} \tag{20}$$

2.3. The damage criterion of TM-DH-BPD model

As with the peridynamics-based damage criterion, once the peridynamic bond strain is greater than its critical value, the peridynamic bond will be damaged. In this way, the damage criterion of the proposed TM-DH-BPD model is expressed as

$$\mu\langle\boldsymbol{\xi}\rangle=\begin{cases}1, s\langle\boldsymbol{\xi}\rangle-s_\theta\langle\boldsymbol{\xi}\rangle\le s_c\langle\boldsymbol{\xi}\rangle\\ 0, s\langle\boldsymbol{\xi}\rangle-s_\theta\langle\boldsymbol{\xi}\rangle> s_c\langle\boldsymbol{\xi}\rangle\end{cases}\tag{21}$$

where $s_\theta\langle\boldsymbol{\xi}\rangle$ denotes the thermal bond strain associated with the bond $\langle\boldsymbol{\xi}\rangle$,$s_\theta=\alpha\cdot\Delta\theta$ (α is the coefficient of thermal expansion). $\mu\langle\boldsymbol{\xi}\rangle$ is the damage index of the peridynamic bond $\langle\boldsymbol{\xi}\rangle$. $\mu\langle\boldsymbol{\xi}\rangle=1$ denotes that the bond $\langle\boldsymbol{\xi}\rangle$ is undamaged. Meanwhile, $\mu\langle\boldsymbol{\xi}\rangle=0$ denotes that the bond $\langle\boldsymbol{\xi}\rangle$ is damaged. $s_c\langle\boldsymbol{\xi}\rangle$ denotes the critical bond strain [56]:

$$s_c=\begin{cases}\sqrt{\dfrac{4\pi G_c}{9E\delta_x}}, & \text{Plane stress}\\ \sqrt{\dfrac{5\pi G_c}{12E\delta_x}}, & \text{Plane strain}\\ \sqrt{\dfrac{5G_c}{6E\delta_x}}, & 3D\end{cases}\tag{22}$$

where G_c denotes the fracture toughness. The fracture toughness G_c can be simply replaced by the interface fracture toughness G_c^I (or called the interface critical energy release rate) to address the interface damage problems.

The local damage index $d(\mathbf{x})$ of the material point $\mathbf{x}$ is given by

$$d(\mathbf{x})=1-\frac{\int_{\mathscr{H}_x}\mu\langle\boldsymbol{\xi}\rangle\mathrm{dV}_{x'}}{\int_{\mathscr{H}_x}\mathrm{dV}_{x'}}\tag{23}$$

2.4. Abaqus implementation of TM-DH-BPD model

The coupled governing equations of the TM-DH-BPD model are

$$\begin{cases}\rho\ddot{u}(x,t)=\displaystyle\int_{\mathscr{H}_x}\underline{T}[x,t]\langle\xi\rangle\mathrm{dV}_{x'}+\int_{\mathscr{H}'_x}\underline{T}[x',t]\langle\xi\rangle\mathrm{dV}_{x'}+b(x,t)\\ \rho\cdot C_v\cdot\dot{\theta}(x,t)=\displaystyle\int_{\mathscr{H}_x}\underline{h}[x,t]\langle\xi\rangle\mathrm{dV}_{x'}+\int_{\mathscr{H}'_x}\underline{h}[x',t]\langle\xi\rangle\mathrm{dV}_{x'}+r_q(x,t)\end{cases}\tag{24}$$

where the dual-horizon peridynamics-based heat flow state $\underline{h}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle$ is given in Eq. (14). When we consider the thermal part, the force state is rewritten as

$$\underline{\mathbf{T}}[\mathbf{x},t]\langle\boldsymbol{\xi}\rangle=\frac{1}{2}c\cdot(s\langle\boldsymbol{\xi}\rangle-s_\theta\langle\boldsymbol{\xi}\rangle)\cdot\frac{\boldsymbol{\xi}+\boldsymbol{\eta}}{|\boldsymbol{\xi}+\boldsymbol{\eta}|}\tag{25}$$

To solve for the peridynamics-based coupled thermo-mechanical equations in Eq. (24), the multi-rate explicit integration approach with Adaptive Dynamic Relaxation (ADR) [43,44] is widely used to deal with the time scale gap of the peridynamics-based coupled thermo-mechanical governing equations. Different from ADR, to guarantee the numerical accuracy of the TM-DH-BPD model, the authors propose the dual-horizon peridynamic finite element method (DHPD-FEM) along with two approaches of the standard implicit solver in Abaqus/Standard and explicit solver with mass scaling in Abaqus/Explicit. Moreover, dual-horizon peridynamic finite element method is implemented in Abaqus/UEL and Abaqus/VUEL to solve for the peridynamics-based coupled thermo-mechanical equations. It can use the powerful computing power and visualization capabilities of Abaqus, which may accelerate the pace of the theoretical researching and real-word application of peridynamics including the complex thermal shock cracking problems of TBCs.

The computational body Ω_0 is firstly discretized by the finite elements (quadrilateral or hexahedral elements) as shown in Fig. 2, and then these finite elements (or called the background elements) are converted into the peridynamic points (or called peridynamic nodes) in Fig. 2. The peridynamic nodes are the center nodes of the correspondent background elements. Additionally, these background elements will be employed to store and visualize the field variable value (including the local damage value, displacement and thermal, et al.) of the peridynamic nodes in real time [55,65,72]. The volume of each peridynamic node is denoted by $\mathrm{V}_i(i=1,2,3,\cdots m)$ such that $\Omega_0=\mathrm{V}_1\cup\mathrm{V}_2\cup\mathrm{V}_3\cup\cdots\cup\mathrm{V}_m$ and $\mathrm{V}_i\cap\mathrm{V}_j=\varnothing, i\neq j$. (Note that V_i is also the volume of the correspondent background element in Fig. 2). Similar to the node-based finite element method [55,65,72], the dual-horizon peridynamic finite element method is provided in this section. For each peridynamic node $\mathbf{x}_i$ (colored red in Fig. 2), the supporting points of the dual-horizon peridynamic element (DH-PDE) E_i consist of those peridynamic nodes (colored green) within the family of $\mathbf{x}_i$ and itself. Then the TM-DH-BPD model can be implemented in Abaqus/UEL for the standard implicit solver and Abaqus/VUEL for the explicit solver. The first work is to obtain the dual-horizon peridynamics-based coupled thermo-mechanical elemental stiffness matrices and residuals in Abaqus/UEL and the dual-horizon peridynamics-based coupled thermo-mechanical elemental internal forces and mass matrices in Abaqus/VUEL.

There are displacement and temperature variables in the reference subregion Ω_0 for dual-horizon peridynamics-based coupled thermo-mechanical problems. Regarding to the dual-horizon peridynamic finite element implementation of the TM-DH-BPD model in Abaqus/UEL, the dual-horizon peridynamics-based coupled thermo-mechanical elemental residual $\mathbf{r}_E=\{\mathbf{r}_E^u,\mathbf{r}_E^\theta\}$ and elemental stiffness matrix $\mathbf{K}_E$ should be given. The dual-horizon peridynamics-based elemental residuals for the displacement and thermal fields can be respectively calculated as

$$\begin{cases}r_E^u=\displaystyle\sum_{x_j\in\mathscr{H}_{x_i}}^{\mathrm{Dual}}\left(\underline{T}[x_i,t]\langle\xi\rangle\mathrm{V}_j\mathrm{V}_i\cdot\eta_{ij}\right)-\sum_{x_i}(\rho\cdot\ddot{u}(x_i)\cdot\mathrm{V}_i)\\ r_E^\theta=\displaystyle\sum_{x_j\in\mathscr{H}_{x_i}}^{\mathrm{Dual}}\left(\underline{h}[x_i,t]\langle\xi\rangle\mathrm{V}_j\mathrm{V}_i\cdot\eta_{ij}\right)-\sum_{x_i}\left(\rho C_v\dot{\theta}(x_i)\cdot\mathrm{V}_i\right)\end{cases}\tag{26}$$

where $\boldsymbol{\xi}=\mathbf{x}_j-\mathbf{x}_i$. $\mathbf{r}_E^u$ and $\mathbf{r}_E^\theta$ denote the dual-horizon peridynamics-based elemental residuals associated with the displacement and thermal fields. The dual assembly symbol $\sum^{\mathrm{Dual}}$ denotes the dual assembly algorithm of the state variable into the peridynamic elemental residual in framework of DH-PD. Additionally, the assembly symbol $\sum$ denotes the ordinary assembly process (the detailed assembly information can be seen in Table 1). The volume correction factor η_{ij} [56] is given by

$$\eta_{ij}=\begin{cases}1 & |\xi|\le(\delta_{x_i}-r_j);\ \ r_j=|\xi|/2\\ \dfrac{\delta_{x_i}+r_j-|\xi|}{2r_j} & (\delta_{x_i}-r_j)<|\xi|\le\delta_{x_i}\\ 0 & |\xi|>\delta_{x_i}\end{cases}\tag{27}$$

where δ_{x_i} denotes the peridynamic horizon of material point $\mathbf{x}_i$. We just modify the equivalent point spacing $\Delta x=\sqrt{V_i/H}$ for 2D, $\Delta x=\sqrt[3]{V_i}$ for 3D to calculate the volume correction factor for non-uniform point spacing. The peridynamic horizon radius is $\delta_{x_i}=m\cdot\Delta x$.

The dual-horizon peridynamics-based elemental stiffness matrices for the displacement and thermal fields are calculated as

$$\begin{cases}K_E^u=\displaystyle\sum_{x_j\in\mathscr{H}_{x_i}}^{\mathrm{Dual}}\underline{\underline{K}}^u[x_i]\cdot\mathrm{V}_j\cdot\mathrm{V}_i\cdot\eta_{ij}+\sum_{x_i}\left(\rho\cdot\mathrm{V}_i\cdot I/\left(\beta_0\cdot\Delta t^2\right)\right)\\ K_E^\theta=\displaystyle\sum_{x_j\in\mathscr{H}_{x_i}}^{\mathrm{Dual}}\underline{\underline{K}}^\theta[x_i]\cdot\mathrm{V}_j\cdot\mathrm{V}_i\cdot\eta_{ij}+\sum_{x_i}(\rho\cdot C_v\cdot\mathrm{V}_i/\Delta t)\end{cases}\tag{28}$$

where Δt denotes the time step. $\mathbf{I}$ denotes the identity matrix. We set

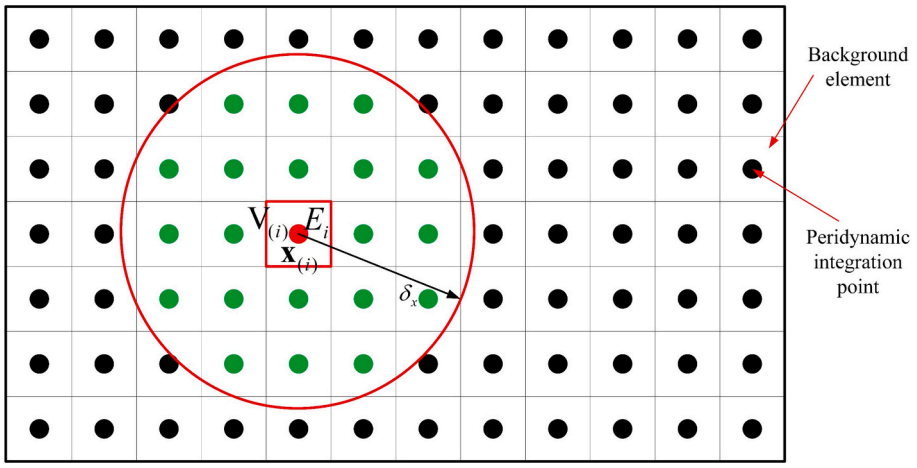


Fig. 2. The computational body Ω_0 discretized by the peridynamic material points.

Table 1
The dual-horizon peridynamics-based elemental stiffness matrices and residuals of the TM-DH-BPD model in Abaqus/UEL.

1: Initialize peridynamics-based elemental stiffness matrices and residuals for DH-PDE E_i
2: $\mathbf{K}_E(1:(d+1)\cdot n_i, 1:(d+1)\cdot n_i) = \mathbf{0}.; \mathbf{K}_E^u(1:d\cdot n_i, 1:d\cdot n_i) = \mathbf{0}; \mathbf{K}_E^\theta(1:n_i, 1:n_i) = \mathbf{0}$.
3: $\mathbf{r}_E(1:(d+1)\cdot n_i, 1) = \mathbf{0}$, $\mathbf{r}_E^u(1:d\cdot n_i, 1) = \mathbf{0}, \mathbf{r}_E^\theta(1:n_i, 1) = \mathbf{0}$
4: n_i denotes the total number of nodes for DH-PDE E_i. $d = 2$ for the two-dimensional cases,
5: $d = 3$ for the three-dimensional cases.
6: **do** $j = 2, n_i$!! $\mathbf{x}_j \in \mathscr{H}_{\mathbf{x}_i}$, $\boldsymbol{\xi} = \mathbf{x}_j - \mathbf{x}_i$.
7: Calculate $\underline{\mathbf{T}}[\mathbf{x}_i, t]$ in Eq. (25) and $\underline{h}[\mathbf{x}_i, t]$ in Eq. (14).
8: Dual assembly of $\underline{\mathbf{T}}[\mathbf{x}_i, t]$ to the peridynamic elemental residual $\mathbf{r}_E^u$ for the displacement sub-problem.
$\mathbf{r}_E^u(1:d, 1) = \mathbf{r}_E^u(1:d, 1) + \underline{\mathbf{T}}[\mathbf{x}_i, t]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$\mathbf{r}_E^u(d\cdot j - d + 1 : d\cdot j, 1) = \mathbf{r}_E^u(d\cdot j - d + 1 : d\cdot j, 1) - \underline{\mathbf{T}}[\mathbf{x}_i, t]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
9: Dual assembly of $\underline{h}[\mathbf{x}_i, t]$ to the peridynamic elemental residual $\mathbf{r}_E^\theta$ for the thermal sub-problem.
$\mathbf{r}_E^\theta(1, 1) = \mathbf{r}_E^\theta(1, 1) + \underline{h}[\mathbf{x}_i, t]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$\mathbf{r}_E^\theta(j, 1) = \mathbf{r}_E^\theta(j, 1) - \underline{h}[\mathbf{x}_i, t]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
10: Calculate $\underline{\underline{\mathbf{K}}}^u[\mathbf{x}_i]$ in Eq. (11) and $\underline{\underline{K}}^\theta[\mathbf{x}_i]$ in Eq. (20).
11: Dual assembly of $\underline{\underline{\mathbf{K}}}^u[\mathbf{x}_i]$ to the peridynamics-based mechanical elemental stiffness matrix $\mathbf{K}_E^u$.
$K_E^u(1:d, 1:d) = K_E^u(1:d, 1:d) + \underline{\underline{K}}^u[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$K_E^u(1:d, d\cdot j - d + 1 : d\cdot j) = K_E^u(1:d, d\cdot j - d + 1 : d\cdot j) - \underline{\underline{K}}^u[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
12: $K_E^u(d\cdot j - d + 1 : d\cdot j, 1:d) = K_E^u(d\cdot j - d + 1 : d\cdot j, 1:d) - \underline{\underline{K}}^u[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$K_E^u(d\cdot j - d + 1 : d\cdot j, d\cdot j - d + 1 : d\cdot j) =$
$K_E^u(d\cdot j - d + 1 : d\cdot j, d\cdot j - d + 1 : d\cdot j) + \underline{\underline{K}}^u[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
13: Dual assembly of $\underline{\underline{K}}^\theta[\mathbf{x}_i]$ to the peridynamics-based thermal elemental stiffness matrix $\mathbf{K}_E^\theta$.
14:
$K_E^\theta(1,1) = K_E^\theta(1,1) + \underline{\underline{K}}^\theta[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$K_E^\theta(1,j) = K_E^\theta(1,j) - \underline{\underline{K}}^\theta[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$K_E^\theta(j,1) = K_E^\theta(j,1) - \underline{\underline{K}}^\theta[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
$K_E^\theta(j,j) = K_E^\theta(j,j) + \underline{\underline{K}}^\theta[x_i]\cdot \mathrm{V}_j\cdot \mathrm{V}_i\cdot\eta_{ij}$
15: **end do**!! $j = 2, n_i$
16: $\mathbf{r}_E(1:d\cdot n_i, 1) = \mathbf{r}_E^u(1:d\cdot n_i, 1); \mathbf{r}_E(d\cdot n_i + 1 : (d+1)\cdot n_i, 1) = \mathbf{r}_E^u(1:n_i, 1)$
17:
$\mathbf{K}_E(1:d\cdot n_i, 1:d\cdot n_i) = \mathbf{K}_E^u(1:d\cdot n_i, 1:d\cdot n_i)$
$\mathbf{K}_E(d\cdot n_i + 1 : (d+1)\cdot n_i, d\cdot n_i + 1 : (d+1)\cdot n_i) = \mathbf{K}_E^\theta(1:n_i, 1:n_i)$
18: If the inertia term is considered, then
19:
$\mathbf{r}_E(1:d, 1) = \mathbf{r}_E(1:d, 1) - \rho\cdot\ddot{\mathbf{u}}(\mathbf{x}_i)\cdot \mathrm{V}_i$
$\mathbf{K}_E(1:d, 1:d) = \mathbf{K}_E(1:d, 1:d) + \rho\cdot \mathrm{V}_i\cdot \mathbf{I}/(\beta_0\cdot\Delta t^2)$
20: If the dynamic heat conduction is considered, then
21:
$\mathbf{r}_E(d\cdot n_i + 1, 1) = \mathbf{r}_E(d\cdot n_i + 1, 1) - \rho C_v\dot{\theta}(\mathbf{x}_i)\cdot \mathrm{V}_i;$
$\mathbf{K}_E(d\cdot n_i + 1, d\cdot n_i + 1) = \mathbf{K}_E(d\cdot n_i + 1, d\cdot n_i + 1) + (\rho\cdot C_v\cdot \mathrm{V}_i)/\Delta t$

$\beta_0 = 0.25$ to guarantee the numerical stability. $\mathbf{K}_E^u$ and $\mathbf{K}_E^\theta$ denote the dual-horizon peridynamic elemental stiffness matrices associated with the displacement and thermal fields, respectively. The dual assembly of double state (or called the bond stiffness) into dual-horizon peridynamic elemental stiffness matrices can be seen in Table 1.

Regarding to the dual-horizon peridynamic finite element implementation of the TM-DH-BPD model in Abaqus/VUEL, the dual-horizon peridynamic elemental internal force vector $\mathbf{f}_E$ and mass matrix $\mathbf{M}_E$ should be given. The specific formulas can be seen in the following parts, meanwhile, their program implementations in Abaqus/VUEL are given in Table 2. The dual-horizon peridynamic elemental internal forces can be calculated as

$$\begin{cases} f_E^u = -\sum\limits_{x_j \in \mathscr{H}_{x_i}}^{\text{Dual}} \underline{T}[x_i, t]\langle\xi\rangle \mathrm{V}_j \mathrm{V}_i\cdot\eta_{ij} \\ f_E^\theta = -\sum\limits_{x_j \in \mathscr{H}_{x_i}}^{\text{Dual}} \underline{h}[x_i, t]\langle\xi\rangle \mathrm{V}_j \mathrm{V}_i\cdot\eta_{ij} \end{cases} \tag{29}$$

where $\mathbf{f}_E^u$ and $\mathbf{f}_E^\theta$ denote the peridynamic elemental internal force vector associated with the displacement and thermal fields, respectively.

Since DHPD-FEM is based on the node-based integration scheme, the integration point is also its peridynamic material point. The mass matrices $\mathbf{M}_E^u$ and $\mathbf{M}_E^\theta$ associated with the displacement and thermal fields are

$$\mathbf{M}_E^u = \rho\cdot \mathrm{V}_i\cdot\mathbf{I} \quad \mathbf{M}_E^\theta = \rho\cdot C_v\cdot \mathrm{V}\cdot\mathbf{I} \tag{30}$$

3. Representative numerical examples

In this section, four representative numerical examples are used to demonstrate the effectiveness of the proposed TM-DH-BPD model and its corresponding numerical method of DHPD-FEM. Note that no parallel computing is used if no otherwise specified. The hardware and software features of the computer are: 11th Gen Intel (R) Core (TM) i9-11900K @ 3.50GHz; RAM: 128 GB. Additionally, since there is no extreme mechanical load (e.g., Impact and collision) in the thermal shock example of this paper, the mechanical deformation-induced changes in heat conduction may be small and generally negligible. As with the representative references [6, 43, 44], the one-way coupling (temperature→stress) is generally adopted to study the thermally induced deformation and fractures of ceramic materials in the context of peridynamics. The numerical results of the references [6, 43, 44] validate that the one-way coupling is effective to handle the thermally induced fractures of ceramic materials. Therefore, the paper just adopts the one-way coupling (temperature→stress) that is consistent with the references

Table 2
The dual-horizon peridynamic elemental internal forces and mass matrices of the TM-DH-BPD model in Abaqus/VUEL.

1: Initialize peridynamics-based internal forces and mass matrices for DH-PDE E_i
2: $\mathbf{f}_E(1:(d+1)\cdot n_i,1)=\mathbf{0}$, $\mathbf{f}_E^u(1:d\cdot n_i,1)=\mathbf{0}, \mathbf{f}_E^\theta(1:n_i,1)=\mathbf{0}$
3: $\mathbf{M}_E(1:(d+1)\cdot n_i,1:(d+1)\cdot n_i)=\mathbf{0}$
4: n_i denotes the total number of nodes for DH-PDE E_i. $d=2$ for the two-dimensional cases,
5: $d=3$ for the three-dimensional cases.
6: Calculate $\mathbf{M}_E^u$ and $\mathbf{M}_E^\theta$ in (30) for DH-PDE E_i
7: $\mathbf{M}_E(1:d,1:d)=\beta_{ms}\mathbf{M}_E^u$; $\mathbf{M}_E(d\cdot n_i+1,d\cdot n_i+1)=\mathbf{M}_E^\theta$; β_{ms} denotes the mass scaling factor.
8: **do** $j=2,n_i$!! $\mathbf{x}_j\in\mathscr{H}_{\mathbf{x}_i}$, $\boldsymbol{\xi}=\mathbf{x}_j-\mathbf{x}_i$.
9: Calculate $\underline{\mathbf{T}}[\mathbf{x}_i,t]$ in Eq. (25) and $\underline{h}[\mathbf{x}_i,t]$ in Eq. (14).
10: Dual assembly of $\underline{\mathbf{T}}[\mathbf{x}_i,t]$ to the peridynamic elemental internal force $\mathbf{f}_E^u$ for the mechanical part.
11: $\mathbf{f}_E^u(1:d,1)=\mathbf{f}_E^u(1:d,1)\ -\underline{\mathbf{T}}[\mathbf{x}_i,t]\cdot\mathbf{V}_j\cdot\mathbf{V}_i\cdot\eta_{ij}$
$\mathbf{f}_E^u(d\cdot j-d+1:d\cdot j,1)=\mathbf{f}_E^u(d\cdot j-d+1:d\cdot j,1)+\underline{\mathbf{T}}[\mathbf{x}_i,t]\cdot\mathbf{V}_j\cdot\mathbf{V}_i\cdot\eta_{ij}$
12: Dual assembly of $\underline{h}[\mathbf{x}_i,t]$ to the peridynamic elemental internal force $\mathbf{f}_E^\theta$ for the thermal part.
13: $\mathbf{f}_E^\theta(1,1)=\mathbf{f}_E^\theta(1,1)\ -\underline{h}[\mathbf{x}_i,t]\cdot\mathbf{V}_j\cdot\mathbf{V}_i\cdot\eta_{ij}$
$\mathbf{f}_E^\theta(j,1)=\mathbf{f}_E^\theta(j,1)+\underline{h}[\mathbf{x}_i,t]\cdot\mathbf{V}_j\cdot\mathbf{V}_i\cdot\eta_{ij}$
14: **end do**!! $j=2,n_i$

[6, 43, 44].

3.1. The numerical accuracy investigation of the TM-DH-BPD model

The numerical accuracy of the TM-DH-BPD model with different forms of the peridynamics-based micro-thermal conductivities will be investigated in this part. And then, we will give its optimum mathematical expression form. To this end, we investigate the deformation and temperature distributions of the square plate that suffers from a thermal shock on the upper surface in Fig. 3. This type of numerical example was usually used to study the accuracy of the coupled thermo-mechanical models [43,44]. As with the reference [43], the upper surface of the square plate is heated by the temperature of $\theta = 274$ K, meanwhile, the initial thermal shock temperature of the plate is $\theta_0 = 273$ K. We assume the plane stress condition in the section.

The material properties of this numerical example are shown in Table 3. It is discretized by the quadrilateral elements with the mesh size of $\Delta x = 0.01$ cm and then converted to 10,000 material points (the number of dual-horizon peridynamic element (DH-PDE) is also 10,000). The peridynamic horizon radius is $\delta_x = 4.015\Delta x$. The analysis step of "Coupled Temperature-displacement" and the response option of "Transient" are chosen in Abaqus/Standard to address the thermally-induced deformation problems. No cracks and inertia term are considered in this section. Since the implicit solver in Abaqus/Standard is adopted, the larger time step of $\Delta t = 0.01$ s can be used if no otherwise specified. The total time of $t = 10$ s is chosen. To investigate the effectiveness of the TM-DH-BPD model, the L_2 error of the temperature is defined as

$$\|\text{Error}\|_{L_2}=\frac{\left\|\theta_{\text{PD}}-\theta_{\text{Abaqus}}\right\|}{\left\|\theta_{\text{Abaqus}}\right\|},\ \|\theta\|=\left(\int_{\Omega_0}\theta\cdot\theta\,\mathrm{d}\Omega_0\right)^{\frac{1}{2}} \tag{31}$$

where θ_{PD} and θ_{Abaqus} denotes the temperatures calculated by the TM-DH-BPD model and Abaqus.

The temperature evolutions at the central node a of square plate in Fig. 3 for different forms of micro-thermal conductivity $k(\mathbf{x},\mathbf{x}')$ are shown in Fig. 4. When the constant micro-thermal conductivity in Eq. (15) is used, the predicted temperatures for the case of $\beta = 2$ is much accurate than that for the case of $\beta = 1$. When the nonlinear micro-thermal conductivity in Eq. (15) is used, things are different. Take $\gamma = 1$ as an example, the predicted temperatures for the case of $\beta = 1$ is much accurate than that for the case of $\beta = 2$. Overall, the model with nonlinear micro-thermal conductivity is much effective than the constant micro-thermal conductivity in predicting the temperature evolution.

To better investigate the influences of the parameters β and γ on the accuracy of the TM-DH-BPD model, the errors in the L_2 temperature norm at the central node a of the heated square plate in the time period of $t = 10$ s are illustrated in Fig. 5. When $\beta = 2$, the error in the L_2 temperature norm decreases with the increase of the shape parameter of micro-conductivity function γ. When $\beta = 0.5$, the error increases with the increase of γ. When $\beta = 1$, the error maintains a minimum value compared with the cases of $\beta = 0.5$ and $\beta = 2$ for a certain γ. Additionally, the error firstly decreases and then increases with the increase of γ, the error for the case of $\beta = 1$, $\gamma = 0.5$ will be the smallest value. Therefore, the optimum parameters of $\beta = 1$ and $\gamma = 0.5$ will be chosen and fixed in the following parts.

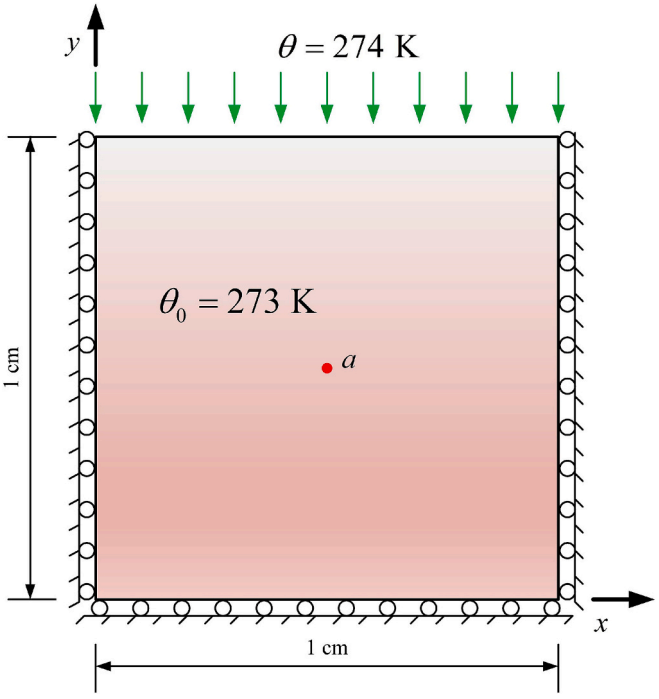


Fig. 3. The square plate under thermal shocks.

Table 3
Material properties of the square plate.

Elastic modulus E (GPa)	1	Conductivity κ (W/(m·K))	1.14
Poisson's ratio ν	1/3	Heat capacity c (J/(kg·K))	100
Density ρ (kg/m^3)	1000	Thermal expansion coefficient α (K^{-1})	1.0E-5

The comparison of the temperature and vertical displacement at the central node a of heated square plate obtained by different peridynamic models with different solvers are given in Fig. 6. The numerical results calculated by the TM-DH-BPD model with the implicit solver of the dual-horizon peridynamic finite element method (DHPD-FEM) converges to that obtained by FEM with the decrease of the time step from $\Delta t = 0.1$ s to $\Delta t = 0.01$ s. Additionally, the numerical results calculated by the TM-DH-BPD model with the implicit solver of DHPD-FEM is more accurate than the TM-BPD model [43] to address the thermally-induced deformation problems in Fig. 6. Since the dual-horizon peridynamic finite element method (DHPD-FEM) is implemented in Abaqus/UEL, the parallel computing can be conveniently realized. To investigate the computational efficiency of the TM-DH-BPD model in Abaqus, the wallclock time varies with CPUs is given in Fig. 7, the calculation time will be reduced by about half when CPUs increase from 1 to 8. Note that the improvement of the parallel efficiency is also related to the computer configuration (that is not the focus of the paper). The purpose of this part is just to show that the parallel computing of the TM-DH-BPD model in Abaqus can be conveniently achieved and improve the computational efficiency.

Note that, although the solver of "Coupled Temperature-displacement" is theoretically suitable to a fully coupled approach in Abaqus, actually, it is also suitable to solve for one-way coupling equation. The numerical results obtained by the one-way coupling with the solver of "Coupled Temperature-displacement" is illustrated in Fig. 6. It shows that the numerical results obtained by the one-way coupling in the context of DHPD-FEM converges to that obtained by Abaqus (fully coupled approach) with the decrease of the time step from $\Delta t = 0.1$ s to $\Delta t = 0.01$ s. Which validates the rationality and effectiveness of one-way coupling and the use of built-in solver of "Coupled Temperature-displacement" of Abaqus in the context of peridynamics. Moreover, the implementation of the proposed DHPD-FEM in Abaqus and using the solver of "Coupled Temperature-displacement" greatly improve the accuracy of the traditional TM-BPD model with the solver of ADR in Fig. 6.

At last, the temperature evolutions and their relative errors at the central node a of heated square plate with different mesh sizes are shown in Fig. 8. The TM-DH-BPD model with the implicit solver of DHPD-FEM can predict the almost same temperature distributions with that obtained by Abaqus in Fig. 8(a). The relative error evolutions of temperature are given in Fig. 8(b). The biggest absolute value of relative error with the relatively bigger mesh size of $\Delta x = 0.01$ cm is less than 0.005%. When the relatively smaller mesh size of $\Delta x = 0.005$ cm is used, the biggest absolute value of relative error is about 0.004%. It means that the relatively bigger mesh size (or point spacing) of the TM-DH-BPD model can yield the convergent results for temperature fields. It demonstrates the effectiveness of the TM-DH-BPD model and its implicit solver of DHPD-FEM.

Remark: The objective of this paper is not to demonstrate that the proposed peridynamics-based model is of better accuracy than traditional PDE-based model. Thanks to the integral form in mathematical formulation and nonlocal interaction in physical interpretation, the obvious advantage of peridynamics-based model may be that it can spontaneously address the crack propagation without extra treatments compared with traditional PDE-based model. We will investigate the thermally induced crack propagation with the proposed TM-DH-BPD model in the following parts. The purpose of the proposed TM-DH-BPD model with the optimum mathematical expression of micro-thermal conductivity is to improve the model accuracy of the traditional peridynamic heat conduction [70,71]. We have given the optimum mathematical expression form by comparing with the numerical predictions of Abaqus in this section. Additionally, the purpose of the proposed DHPD-FEM is to improve the computational accuracy of traditional peridynamic heat conduction with the solver of Adaptive Dynamic Relaxation (ADR) [43]. Moreover, the TM-DH-BPD model with the implicit solver of DHPD-FEM can predict the almost same temperature and displacement distributions with that obtained by Abaqus. Which validates the accuracy and superiority of the TM-DH-BPD model with the implicit solver of DHPD-FEM compared with the traditional TM-BPD model with the implicit solver of ADR. Most importantly, the paper has firstly implemented the peridynamics-based coupled thermo-mechanical model in Abaqus UEL and VUEL subroutines. The implementation of the TM-DH-BPD model in Abaqus can naturally use the built-in powerful solver and post processing capabilities of Abaqus. The parallel computing can be easily realized. All of which are the important contributions in this paper.

3.2. Dynamic crack branching

We will investigate the effectiveness of the TM-DH-BPD model using the example of dynamic crack branching. The thermal part is not considered in this example. The tensile stress σ_0 is suddenly applied on the outer surfaces of the pre-cracked brittle plate in Fig. 9. The numerical example of dynamic crack propagation is very famous for validating the numerical models, e.g. peridynamics [54,56] and phase field models [73–75]. The material properties are copied from the reference [56]. Elastic modulus:$E = 72$ GPa, mass density:$\rho = 2235$ kg/m^3 and the fracture toughness:$G_c = 3.8$ N/m. The plane stress problem is assumed. The TM-DH-BPD model is implemented and calculated in Abaqus/VUEL

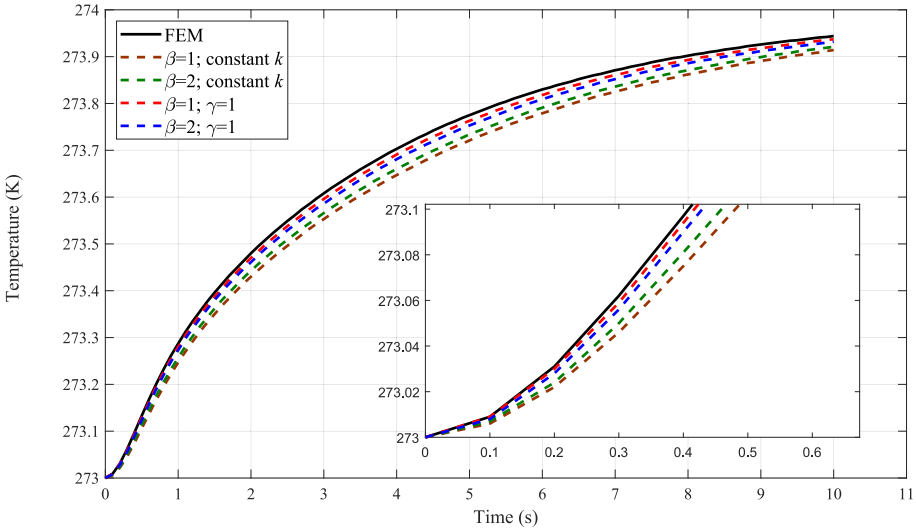


Fig. 4. The temperature evolutions for different forms of micro-thermal conductivity.

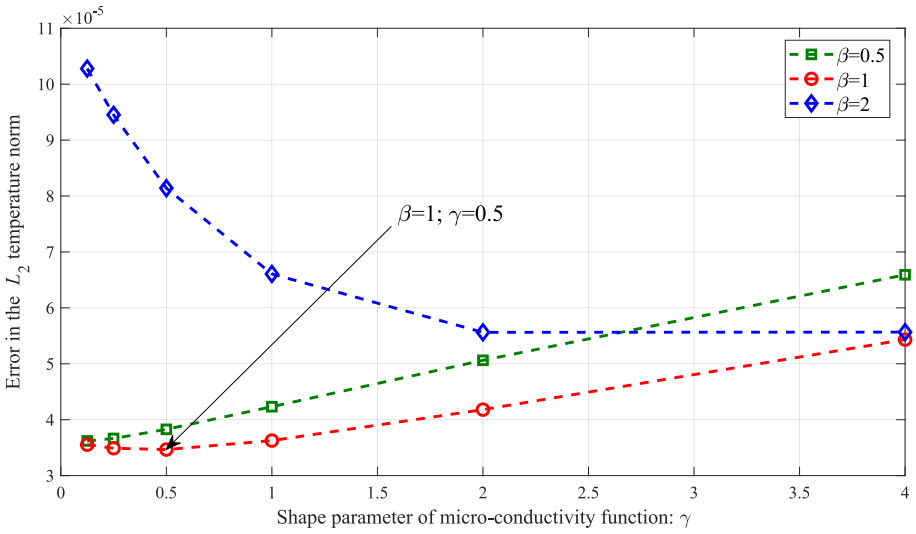


Fig. 5. The errors in the L_2 temperature norm for different parameters β and γ.

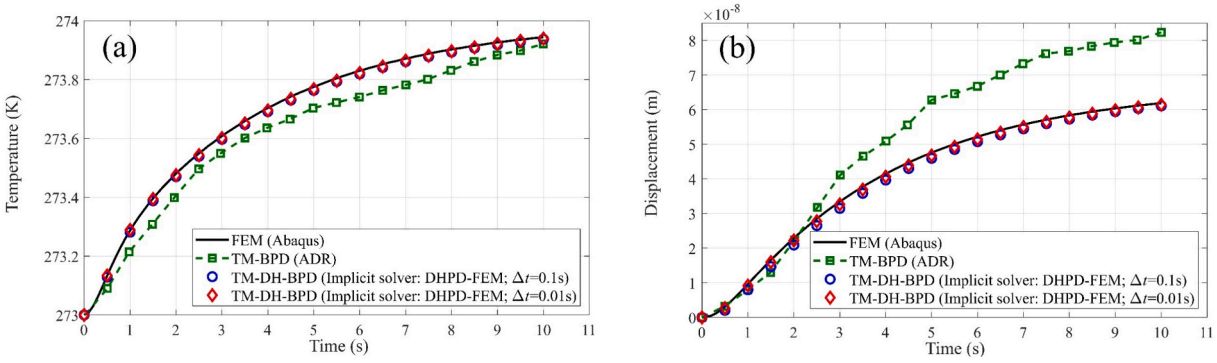


Fig. 6. The comparison of predicted temperature and vertical displacement calculated by the TM-DH-BPD model with the implicit solver of the proposed DHPD-FEM and TM-BPD model with the implicit solver of ADR [43].

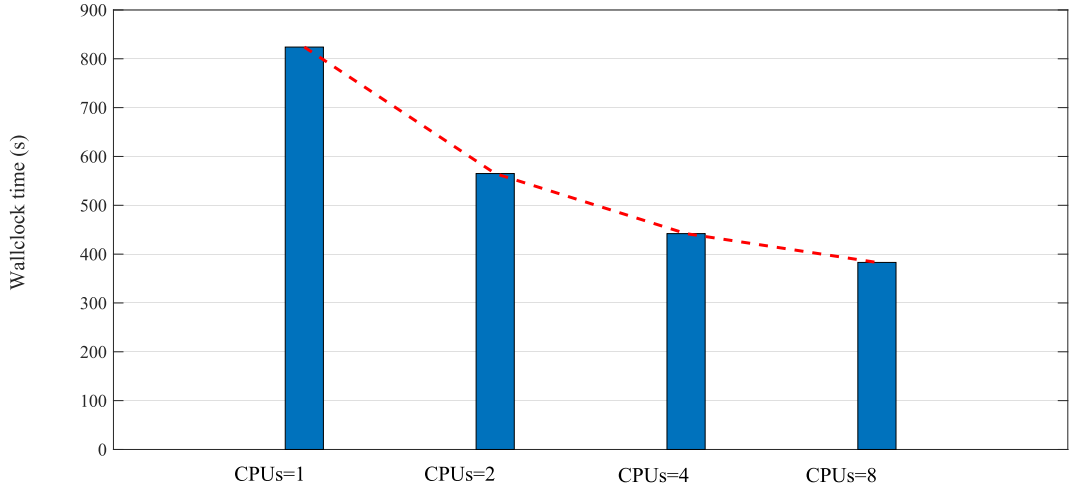


Fig. 7. The wallclock time varies with CPUs with the mesh size of $\Delta x = 0.01$ cm and 100 analysis time steps.

with the explicit solver of the proposed DHPD-FEM. The square plate is firstly discretized by 64,000 uniform quadrilateral elements, then converted to 64,000 material points (the number of dual-horizon peridynamic element is also 64,000). The peridynamic horizon radius is $\delta_x = 4.015\Delta x$. The analysis step of "Dynamic, Explicit" is chosen in Abaqus/Explicit to address the dynamic crack propagation and branching. The time step is set as $\Delta t = 20$ ns.

The evolutions of the displacement and damage fields for the applied stress of $\sigma_0 = 2$ MPa for different output steps are given in Fig. 10 (The calculated results are outputted and visualized for every 0.25 ms).

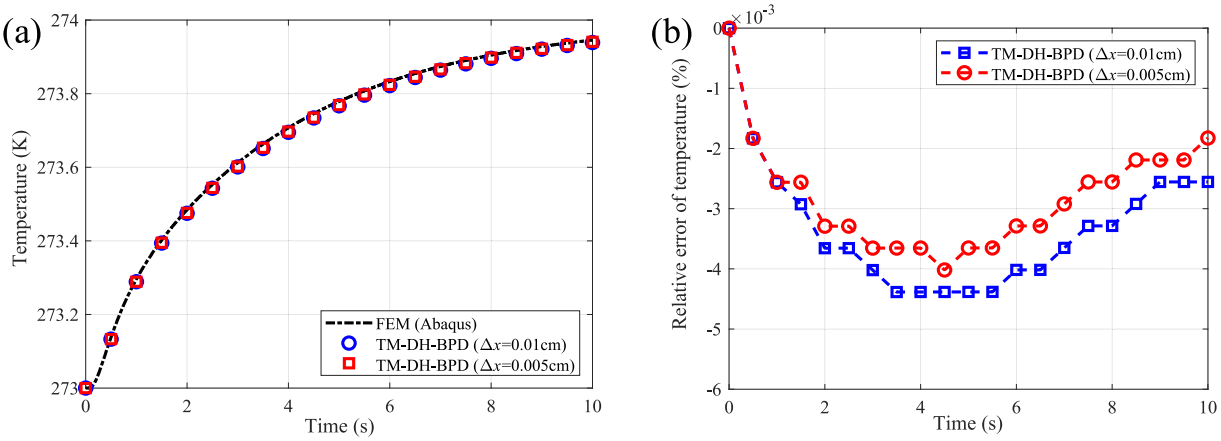


Fig. 8. The convergence investigation for the evolution of temperature field.

SDV_U_Magnitude in the color bar of Fig. 10 denotes the state variable of the displacement magnitude that can be visualized in the post processing module of Abaqus. SDV_Dmg in the color bar denotes the state variable of damage field of the plate. When the output step is 12 in Fig. 10(a), the displacement waves are propagating from the upper and lower sides to its middle place. When the displacement waves reach the initial crack of the plate, it will drive the initial crack propagation and branching from Fig. 10(e-h). To give more details, the video about the displacement and damage evolutions for the applied stress of $\sigma_0 = 2$ MPa is provided in Appendix B (Video B.1). Additionally, the damage evolutions for three different applied stresses are compared in Fig. 11. The dynamic crack propagation patterns obtained by the DH-BPD model with the explicit solver of DHPD-FEM in Abaqus/VUEL are consistent with that calculated by the BPD model [56]. It demonstrates that the implementation of DH-BPD model and its corresponding explicit solver of DHPD-FEM in Abaqus/VUEL is effective to address the dynamic crack propagation and branching problems.

To investigate the convergence and computational efficiency of the proposed model, the damage patterns with different mesh sizes are illustrated in Fig. 12. It can be seen that the width of the crack reduces with the mesh size refining from $\Delta x = 0.5$ mm to $\Delta x = 0.125$ mm. Importantly, the patterns of damage propagation and branching are very similar to each other when using different mesh sizes. To provide more details about the dynamic crack propagation, the crack tip propagation speed profiles (after branching the upper branch is followed) with different mesh sizes are given in Fig. 13. With the mesh size refining, the crack tip propagation speed profile is gradually convergent. The crack tip propagation speed profile with the mesh size of $\Delta x = 0.25$ mm is very consistent with that with the mesh size of $\Delta x = 0.125$ mm. The crack initiation time is $t = 5\,\mu$s. The crack branching time is $t = 20\,\mu$s. Therefore, the current discretization density of $\Delta x = 0.25$ mm yields convergent results for crack paths. Although it is better to reveal the damage propagation and branching when using the relatively smaller mesh size, the computational efficiency and memory usage should be considered. To provide readers with more information, the computational efficiency and memory usage with different mesh sizes and different numbers of CPUs are given in Table 4. The time increment step is fixed as $\Delta t = 10$ ns with different mesh sizes for convenience of comparison. With the mesh size reducing from $\Delta x = 0.5$ mm to $\Delta x = 0.125$ mm, the total numbers of the peridynamic elements increase by 16 times. Meanwhile, the total wallclock time and memory usage for the case of CPUs = 1 increases by 16.53 times and 15.44 times, respectively. When the parallel computing for the case of CPUs = 8 is used, the total wallclock time greatly reduces. Importantly, the memory usage increasing ratio reduces from 1.6358 to 1.1667 with the mesh size reducing from $\Delta x = 0.5$ mm to $\Delta x = 0.125$ mm. At the same time, the speedup ratio increases from 2.5161 to 3.3136. From the above analyses, it demonstrates the convergence of the proposed model when addressing the dynamic crack problems. Thanks to the proposed numerical method of DHPD-FEM and its implementation in Abaqus, it can naturally use the built-in powerful solver and post processing capabilities of Abaqus. This parallel computing can greatly improve the calculation efficiency of the peridynamics-based model without extra treatments. All of which are the important contributions of our paper.

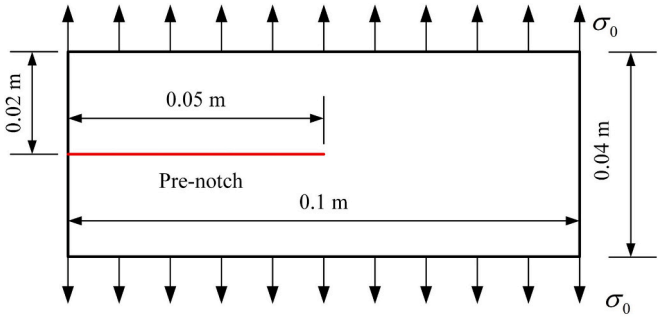


Fig. 9. The brittle plate under an applied tensile stress.

3.3. Thermal shock test of ceramic specimen

The TM-DH-BPD model is applied to address the thermal shock induced damage propagation of the ceramic specimen. In this thermal shock experiment [76], the rectangle ceramic plates with the dimension of 50 mm × 10 mm × 1 mm are heated to different reference temperature θ_0, and then cooled down to $\theta^* = 20^\circ$C. Four ceramic plates of different heated temperatures are studied in the thermal shock test. Because of the symmetric geometry of the plate, the lower left part of it is considered for the calculation efficiency. Thus, the numerical model (25 mm × 5 mm) is shown in Fig. 14. As with the reference [77], the thermal shock temperature is applied on the material points of the left and lower edges. The temperature-independent material proprieties in Table 5 are copied from reference [43]. The plane stress is assumed. As with the assumption [77] that the cracks do not affect the heat transfer, therefore, the breakage of the mechanical bond is assumed not to affect the heat transfer of the thermal bond [77].

We discretize the ceramic plate using the quadrilateral mesh size of $\Delta x = 0.0625$ mm and then convert them to 32,000 material points (the number of dual-horizon peridynamic element is also 32,000). The peridynamic horizon radius is $\delta_x = 4.015\Delta x$. The analysis step of "Dynamic Temperature-displacement, Explicit" is chosen in Abaqus/Explicit to solve the thermally-induced crack problems. Since the explicit solver of Abaqus/VUEL is adopted, to reduce the time scale gap of the coupled

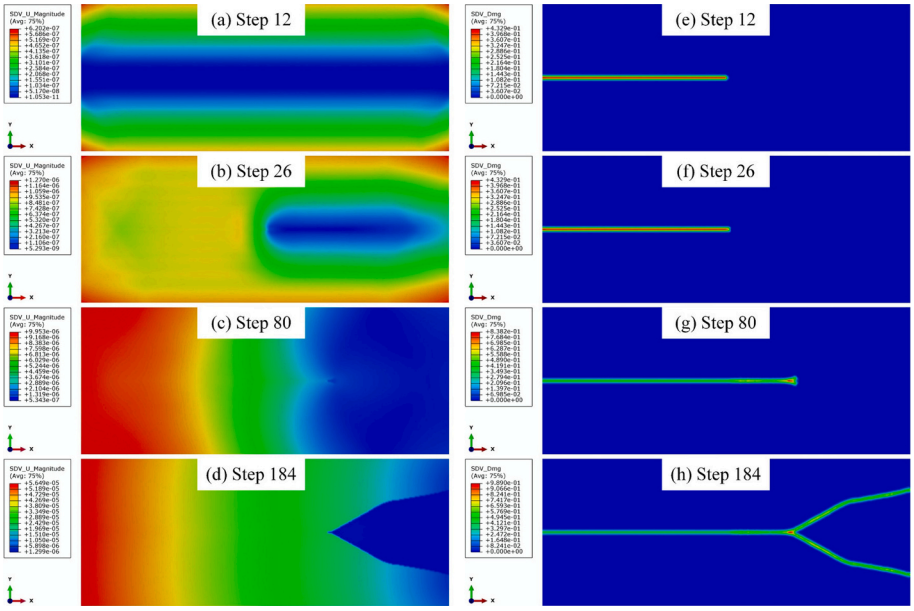


Fig. 10. The evolutions of displacement and damage fields at different output steps, unit: m.

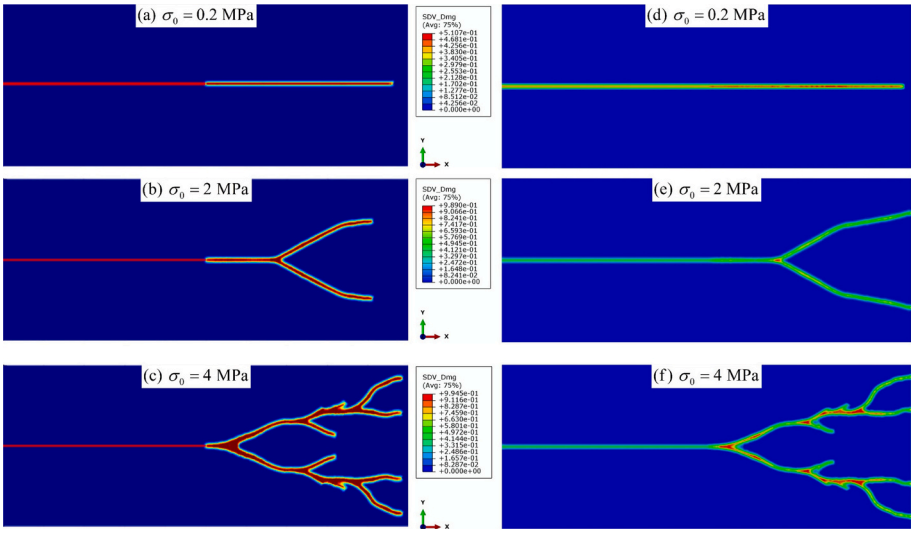


Fig. 11. The damage evolutions for different applied stresses. Left: obtained by BPD model [56]. Right: obtained by DH-BPD model with the explicit solver of DHPD-FEM in Abaqus/VUEL.

thermo-mechanical governing equations, the paper uses the mass scaling method in Abaqus/Explicit. The mass scaling factor is set as 1.0E6 in this example. Since the explicit solver is used, the calculational efficiency and convergence can be greatly improved compared with the implicit solver. The fixed time step of $\Delta t = 5.0\ \mu s$ and total time of $t = 0.3$ s are chosen in this example. The calculated results are outputted and visualized in the post processing module of Abaqus at every 1.0 ms, therefore, the time of the initial output is 1.0 ms.

The evolutions of the total kinetic energy and strain energy of the plate are shown in Fig. 15. When the initial heated temperature is higher, the strain energy is greater in Fig. 15. Additionally, the artificial kinetic energy due to mass scaling remains a very small value for different heated temperatures. It demonstrates the effectiveness of using the mass scaling to increase the simulation time step. With the thermal shock processing, the strain energy of the plate firstly increases to its top value before the initiation of cracks. And then the strain energy slowly decreases when the thermally induced cracks are propagating. The evolutions of thermal and damage fields for the initial temperature of 600 °C at different output steps are shown in Fig. 16. The thermally-induced damages are observed on the thermal shock surfaces in Fig. 16(a). With the time processing, the cooling temperature is transferring from the thermal shock surface of the plate to its inner part in Fig. 16(a-d). The TM-DH-BPD model can capture the periodicity and hierarchy of the fracture patterns in Fig. 16(e-f). Some small cracks propagate and become the large cracks, meanwhile, the growth of other small cracks is inhibited. To provide more detailed information about the crack propagation and the thermal diffusion, the post processing video is given in Appendix B (Video B.2). Additionally, the final thermal crack patterns for different thermal sock temperatures are illustrated in Fig. 17. The predicted fracture patterns are consistent with the experiment observations [76]. It demonstrates that the proposed TM-DH-BPD model and its corresponding dual-horizon peridynamic finite element method in Abaqus/VUEL are effective to address the complex thermal shock fracture patterns.

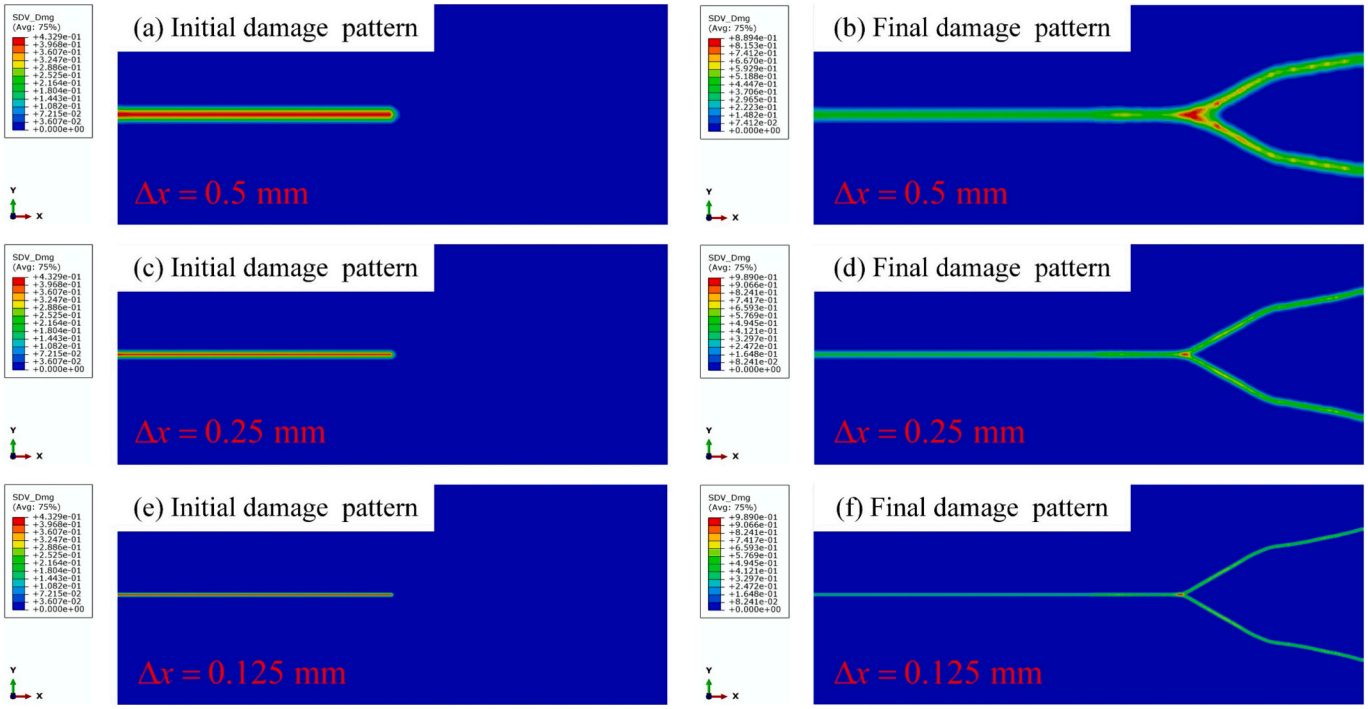


Fig. 12. The initial and final damage patterns with different mesh sizes.

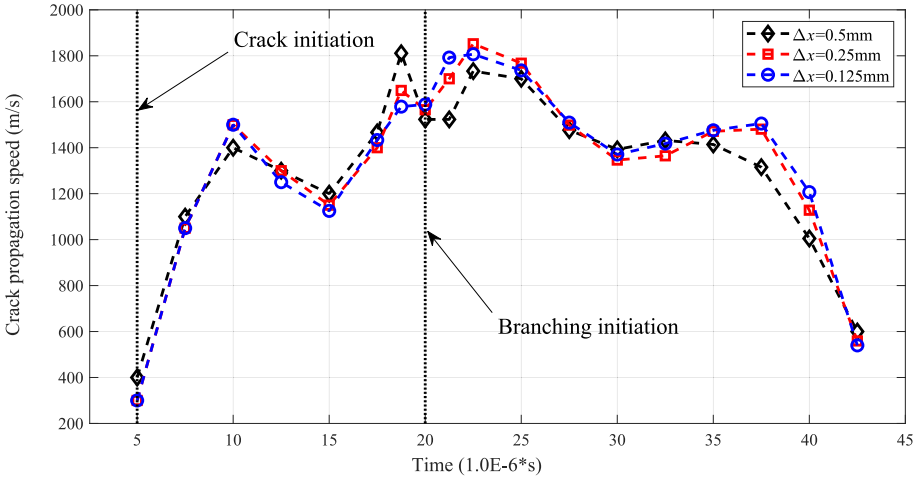


Fig. 13. Comparison of the crack tip propagation speed with different mesh sizes.

It should be noted that the explicit solver with mass scaling in Abaqus have been firstly employed to address the thermally-induced crack problems in the context of peridynamics. The simulation time of quenching test for the TM-DH-BPD model using the explicit solver with mass scaling is about 2.5 h. Since there are 60,000 time increment steps in this example, the computation time per time increment step is about 0.15 s. From this point, it may be of significance in terms of the computational efficiency of the TM-DH-BPD model in Abaqus along with the approach of the explicit solver with mass scaling to address the thermally-induced damage problems.

3.4. *The complex thermal-induced fractures in thermal barrier coatings*

At last, the TM-DH-BPD model is applied to investigate the thermally-induced damage and interface debonding mechanism of TBCs. Liang et al. [3] carried out the thermal shock tests to investigate the influence of the top coat thickness on the cracking behavior in Fig. 18(a-c). The experiment observations show that the interface cracking always occurs for different coating thickness during the thermal shock. Zhao et al. [7] provided the thermal shock experiment results that the interaction phenomenon of multiple transverse cracking and TC/BC interface cracking can be seen in Fig. 18(d).

To explain the causes of different fracture modes observed in the thermal shock experiment of TBCs, we will investigate the thermally induced fracture mechanism in the cases of different TC/BC interface fracture toughness, coating thickness and thermal shock action surface. The 2D and 3D model will be considered in the following parts. We assume the plane stress in two-dimensional numerical example. The 2D and 3D numerical models of TBCs are displayed in Fig. 19. The thickness of BC layer is fixed as $h_{BC} = 100\,\mu$m, meanwhile, the length of TBCs is fixed as $L = 2500\,\mu$m. Since the thermal shock of TBCs mainly cause the inner damages in the TC layer and the interface damages at the TC/BC interface, the substrate is generally ignored for the computational efficiency.

Table 4
The computation time and memory usage with different mesh sizes.

Mesh size	$\Delta x = 0.5$ mm	$\Delta x = 0.25$ mm	$\Delta x = 0.125$ mm
Total peridynamic elements	16,000	64,000	256,000
Total wallclock time (CPUs = 1)	78 s	339 s	1289 s
Total wallclock time (CPUs = 8)	31 s	107 s	389 s
Speedup ratio	78/31 = 2.5161	339/107 = 3.1682	1289/389 = 3.3136
Wallclock time per time increment step (CPUs = 1)	0.0170 s	0.0737 s	0.2802 s
Wallclock time per time increment step (CPUs = 8)	0.0067 s	0.0233 s	0.0846 s
Memory usage (CPUs = 1)	77.7 megabytes	307.6 megabytes	1.2 gigabytes
Memory usage (CPUs = 8)	127.1 megabytes	412.6 megabytes.	1.4 gigabytes
Memory usage increasing ratio	127.1/77.7 = 1.6358	412.6/307.6 = 1.3414	1.4/1.2 = 1.1667

TBCs are discretized by the quadrilateral elements for two-dimensional numerical model (or hexahedral elements for three-dimensional numerical model) with the mesh size of $\Delta x = 5\ \mu$mand then converted to the peridynamic material points. The peridynamic horizon radius is $\delta_x = 4.015\Delta x$. The analysis step of "Dynamic Temperature-displacement, Explicit" is chosen in Abaqus/Explicit to address the thermally-induced crack problems. The mass scaling factor is fixed as 1.0E6 in Abaqus/Explicit. The temperature-independent material parameters can be seen in Table 6. The interface oxidation (TGO growth) is not considered in this example. The time step of $\Delta t = 1.5\ \mu$s and total time of $t = 0.3$ s are set if no otherwise specified.

Note that the material properties of TBCs under high-temperature environments is very complex. Therefore, it is very challenging work to accurately investigate the deformation and damage of TBCs with the consideration of oxidation of the bond coat, creep effects and sintering of the top coat under high-temperature environments. Since no complete temperature-dependent material properties of TBCs (elastic modulus, thermal conductivity, fracture toughness in the bulk material and the interface fracture toughness at the interface) have been provided in the experiments of the literatures, as with many published models [3,6,44],

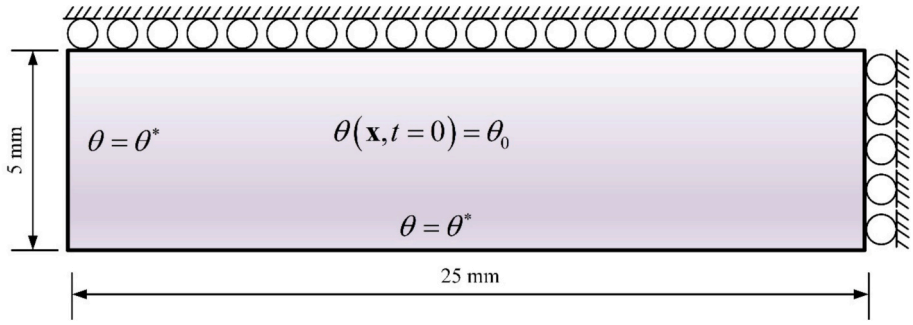


Fig. 14. The quarter of the ceramic plate with the symmetric geometry and boundary conditions.

Table 5
Material proprieties for the thermal shock plate.

Elastic modulus E (GPa)	370	Conductivity κ (W/(m·°C))	31
Fracture toughness G_c (N/m)	42.47	Heat capacity C_v (N·m/(kg·°C))	880
Density ρ (kg/m^3)	3980	Thermal expansion coefficient α (°C^{-1})	7.5E-6

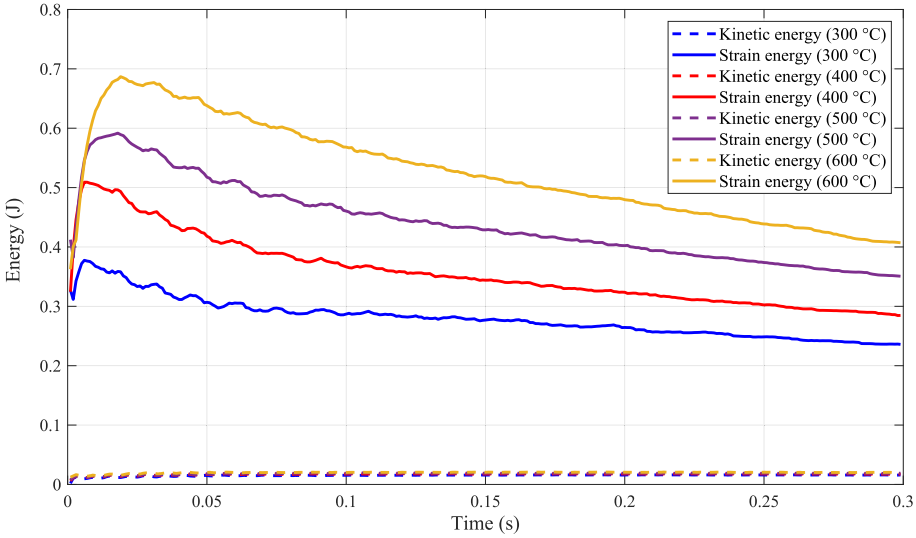


Fig. 15. The comparison of the total kinetic energy and strain energy for different heated temperatures.

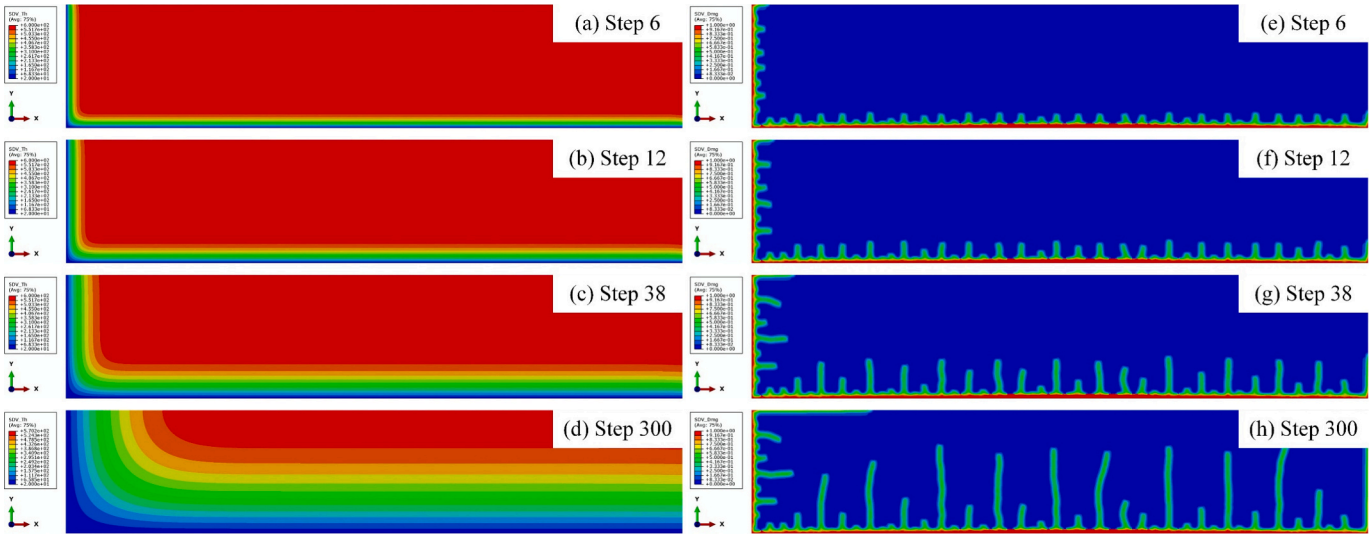


Fig. 16. The evolutions of thermal and damage fields for the initial temperature of 600 °C at different output steps, unit: °C. Left: the thermal evolutions, SDV_Th in the color bar denotes the state variable of thermal field. Right: the damage evolutions of the ceramic specimen.

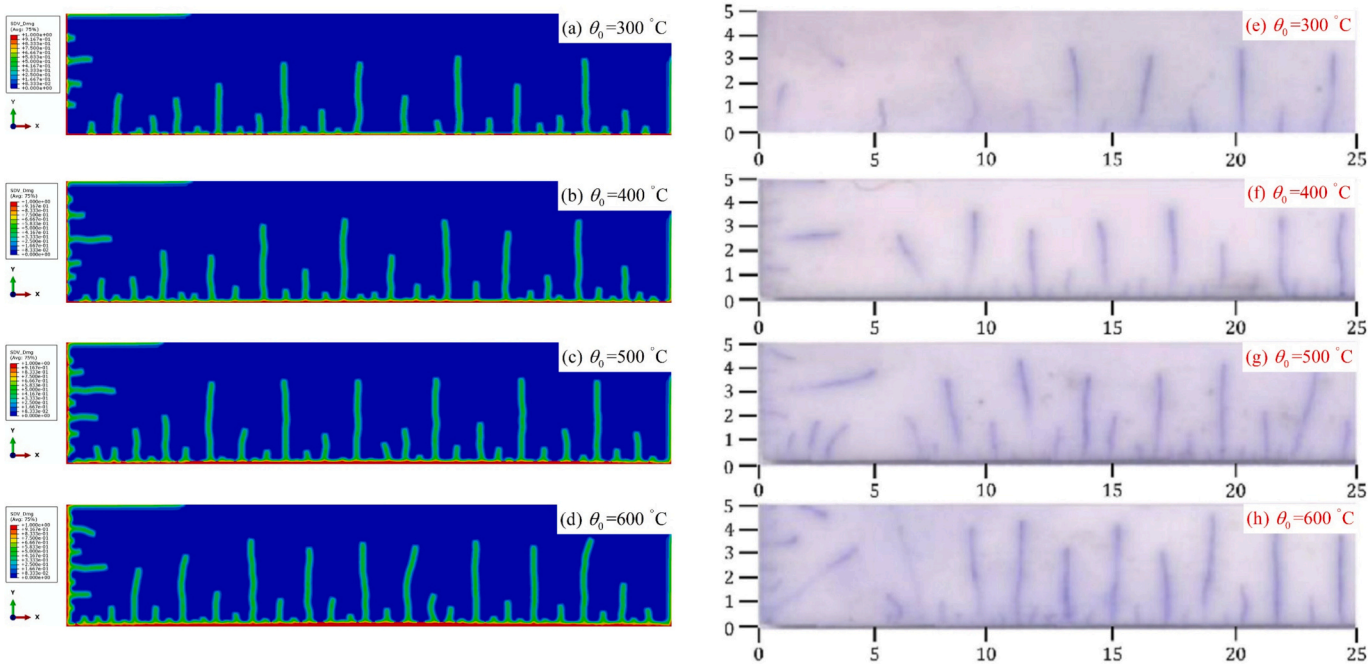


Fig. 17. The damage evolutions for different initial temperatures. Left: obtained by TM-DH-BPD. Right: the experiment results [76].

the temperature-independent material proprieties are assumed to preliminarily reveal the key fracture phenomena observed in the experiment of TBCs. Additionally, this example considers the thermal shock during short time in this section, therefore, the creep deformation of TBCs under long-term high temperatures may be generally neglected. As the future work, we will investigate the thermally-induced damage propagation and interface debonding under temperature cycling in the future. In that case, creep deformation of TBCs and temperature-dependent material properties of TBCs will be considered in the future.

3.4.1. *The damage evolution of TBCs during quenching and heating*

The damage mechanisms of TBCs during quenching and heating are investigated in this section. The thickness of TC layer is fixed as $h_{BC} = 100\ \mu$m in this example. TBCs are firstly heated to $\theta_0 = 1000$°C and rapidly dropped into water at a temperature of $\theta^* = 10$°C [3], this process is called the quenching of thermal shock. TBCs are heated from the initial temperature of $\theta_0 = 10$°C to $\theta^* = 1000$°C, this process is called the heating of thermal shock. When TBCs are in actual service, the upper surface of ceramic layer suffers from the thermal shock. However, the thermal shock may be applied on all the surface of TBCs in the experiment [3] (or the upper and lower surfaces of the TC layer in the experiment [7]). The influence of thermal shock surface on the cracking behaviors of TBCs will be studied in the next sections. The thermal shock is applied on the upper surface of TC layer in this part. The TC/BC interface fracture toughness is fixed as $G_c^{TC/BC} = 0.001$ N/mm in this section.

The damage evolution of TBCs during the quenching process is shown in Fig. 20(a). The multiple transverse cracking can be observed on the upper surface of TBCs at the time of $t = 1$ ms. With the progress of thermal shock, some small transverse cracks propagate and become the

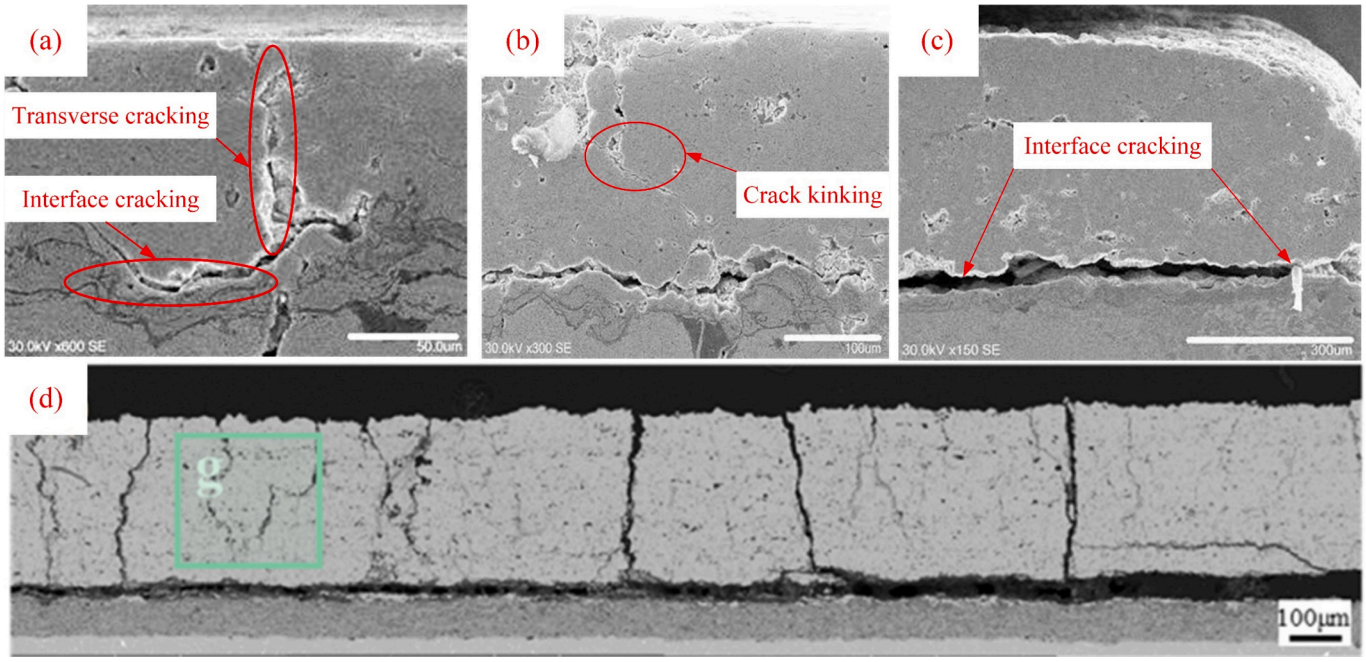


Fig. 18. The thermal shock experiments. (a) The damage mode for coating thickness of $h_{TC} = 100\,\mu$m [3]. (b) The damage mode for coating thickness of $h_{TC} = 300\,\mu$m [3]. (c) The damage mode for coating thickness of$h_{TC} = 300\,\mu$m [3]. (d) The interaction phenomenon of multiple transverse cracking and interface cracking [7].

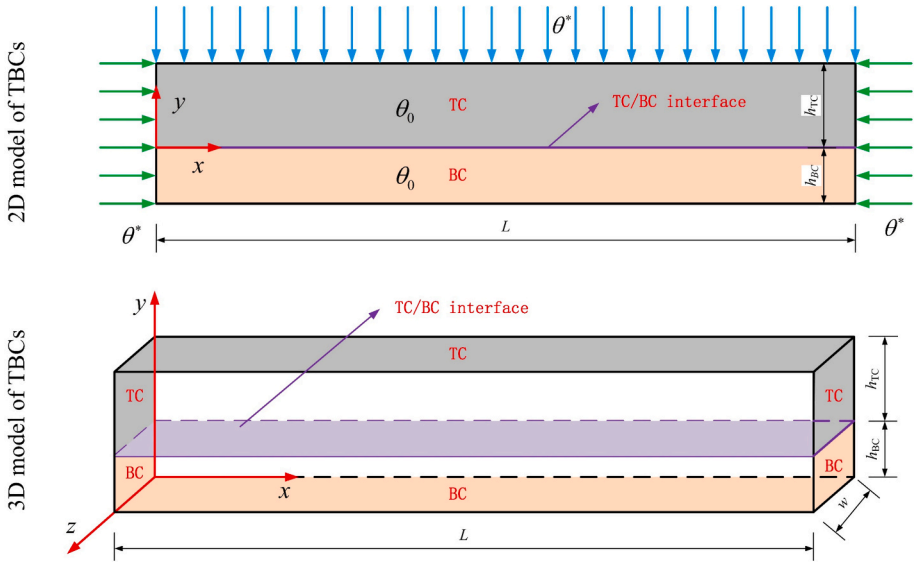


Fig. 19. The 2D and 3D numerical models of TBCs. h_{TC} and h_{BC} denote the thickness of TC and BC layers, respectively. w denotes the width of 3D model in the z direction. L denotes the length of the numerical model.

Table 6
Material properties of TBCs.

Properties	TC	BC
Elastic modulus E (GPa)	11.7	71.9
Thermal expansion coefficient α $\left(\times 10^{-6}{}^{\circ}\mathrm{C}^{-1}\right)$	10.04	17.65
Density ρ $\left(\mathrm{kg/m^3}\right)$	5650	7320
Conductivity κ $\left(\mathrm{W/(m\cdot{}^{\circ}C)}\right)$	1.05	4.3
Heat capacity C_v $\left(\mathrm{N\cdot m/(kg\cdot{}^{\circ}C)}\right)$	450	780
Fracture toughness G_c(N/mm)	0.008	0.3

large transverse cracks. Meanwhile, the growth of other small transverse cracks is inhibited. At the time of $t = 2$ ms in Fig. 20(a), two TC/BC interface cracks propagate from the left and right ends of TBCs to the middle. At the time of $t = 3$ ms in Fig. 20(a), the transverse cracks in the middle of TBCs propagate at the interface of TC/BC layers and cause new interface cracks. The interaction of the transverse damages and the interface damages may cause serious spalling failure of TBCs. At the time of $t = 28$ ms, one can see the complete spalling failure of TC/BC interface. It can be seen that the TM-DH-BPD model can capture the interaction phenomenon of multiple transverse cracking and interface cracking observed in experiment (See Fig. 18(a) and (d)). The total dual-horizon peridynamic element is 20,000. The computational time per analysis increment step and total memory usage for the 2D TBC model

are 0.11 s and 141.1 megabytes, respectively. To provide more detailed information about the damage evolutions and interaction under thermal shock of quenching, the post processing video is given in Appendix B (Video B.3). The damage evolution of TBCs during the heating process is given in Fig. 20(b). Expect for the damage on the upper surface of ceramic layer, it mainly causes the spalling failure of TC/BC interface. However, the mechanism of the spalling failure during heating is different from that during quenching. The interface damage firstly accumulates from the heating time of $t = 8$ ms to $t = 10$ ms. When the heating time is $t = 12$ ms, the obvious spalling failure of TC/BC interface suddenly occurs. The interface is completely damaged until the heating time is $t = 20$ ms.

3.4.2. *The fracture behavior for different TC/BC interface fracture toughness*

The effect of TC/BC interface property on the fracture behavior of TBCs is investigated in this section. TC/BC interface property ranges from "weak" interface to "strong" interface in such a way that the interface fracture toughness $G_c^{TC/BC}$ is set as 0.001 N/mm, 0.002 N/mm, 0.004 N/mm and 0.008 N/mm. The thermal shock is applied on the upper surface of TC layer. The total time is $t = 0.3$ s. The final damage patterns of TBCs for different interface fracture toughness during quenching are given in Fig. 21. When the interface fracture toughness increases from 0.001 N/mm to 0.004 N/mm in Fig. 21 (a-c), the final damage patterns are almost the same with each other. When the TC/BC interface is relatively strong ($G_c^{TC/BC} = 0.008$ N/mm), the TC/BC interface is not fully damaged. Additionally, there is no interaction phenomenon of multiple transverse cracking and interface cracking. It can be seen that the quenching process of TBCs mainly causes multiple transverse cracking when the interface is strong. The interaction of the transverse and interface cracks can be observed for the relatively weak interface, which may accelerate the spalling failure. The multiple transverse cracks mainly exist for the relatively strong interface during the quenching process. Whether or not the transverse and interface cracks interact is determined by TC/BC interface property.

It can be seen from the thermal shock experiment of TBCs [3,7] (or see Fig. 18) that the transverse cracking propagation, kinking in the top coat and its interaction with the TC/BC (Top coat/Bond coat) interfacial cracking can be observed, there is no obvious damage in the BC layer during the thermal shock of TBCs. Which means that the transverse cracks will not propagate through the TC/BC interface and propagate into the BC layer in the thermal shock experiment of TBCs. As the same with the experimental observations [3,7], the numerical results obtained by the proposed TM-DH-BPD model show that the transverse crack will only deflect along the TC/BC interface (deflection).

To further explain why the transverse crack will only deflect along the interface from the theoretical perspective, the theoretical predictions from He and Hutchinson [78] are used to validate the simulation results. The ratio R between the fracture toughness for a crack deflecting into the interface and the fracture toughness for a crack penetrating into the bulk is determined by the included angle β_i between the crack propagation direction and the interface direction [78]:

$$R = \frac{1}{16}\left[\left(3\cos\frac{\beta_i}{2} + \cos\frac{3\beta_i}{2}\right)^2 + \left(\sin\frac{\beta_i}{2} + \sin\frac{3\beta_i}{2}\right)^2\right] \tag{32}$$

When $R > G_c^i/G_c^b$ (G_c^i denotes the critical fracture toughness; G_c^b denotes the critical bulk fracture toughness), the crack deflects into the interface. Otherwise, the crack penetrates into the bulk. Regarding to the TBC thermal shock in this section, the included angle between the transverse crack direction and TC/TC interface is $\beta_i = 90°$, therefore, the critical fracture toughness ratio for crack deflection/penetration $R = 0.25$ from Eq. (32). The fracture toughness of BC material in Table 6 is $G_c^b = 0.3$ N/mm, therefore, the critical interface fracture toughness is $G_c^i = G_c^b \cdot R = 0.075$ N/mm. This section investigates the thermal shock responses of thermal barrier coatings under different interface fracture toughness values ($G_c^{TC/BC} = 0.001$ N/mm – 0.008 N/mm). Since the interface fracture toughness values $G_c^{TC/BC}$ is lower than the critical interface fracture toughness $G_c^i = 0.075$ N/mm, therefore, the transverse cracks deflect into the TC/BC interface. The numerical results in this section show that the transverse crack will only deflect into the TC/BC interface, which is consistent with the experiment data in Fig. 18 and theoretical predictions in Eq. (32).

3.4.3. *The fracture behavior for different coating thickness*

And then, we investigate the influence of the coating thickness on the

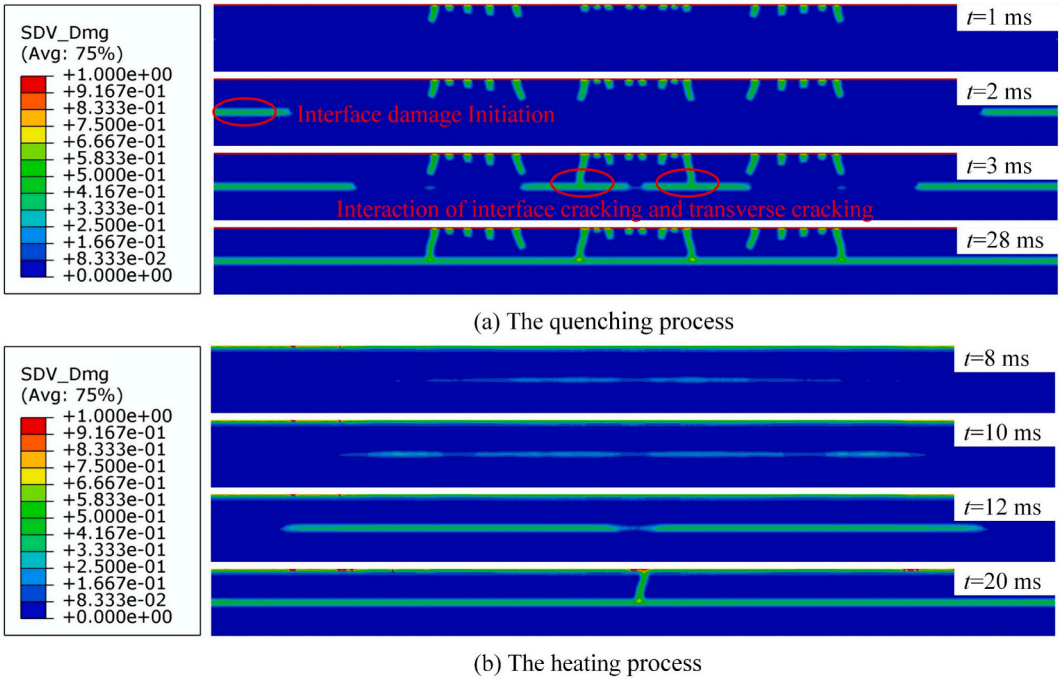


Fig. 20. The comparison of the damage evolution of TBCs during quenching and heating.

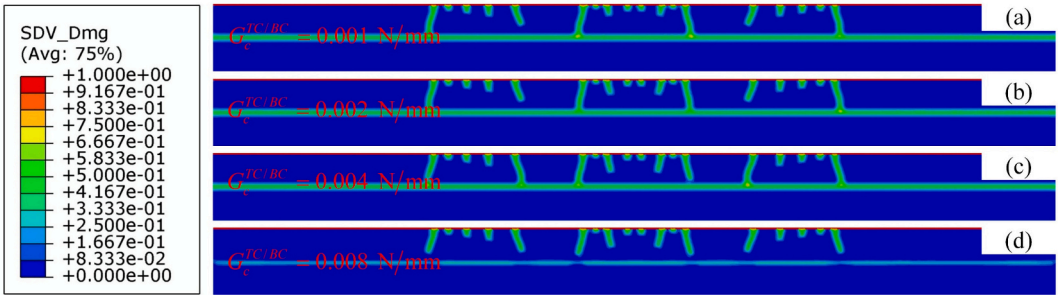


Fig. 21. The final damage patterns of TBCs for different interface fracture toughness during quenching.

cracking behavior of TBCs. The thermal shock is also applied on the upper surface of TC layer. The weak interface is assumed ($G_c^{TC/BC} = 0.001$ N/mm). The coating thicknesses are set as three different values,$h_{TC} = 100\ \mu$m, 300 μm, 500 μm. The final damage patterns of TBCs are displayed in Fig. 22. The spalling failure of TC/BC interface can be seen for different coating thicknesses, which is similar to the experiment results of Fig. 18(a-c). When the thickness of the TC layer increases, the phenomenon of multiple transverse crack kinking that observed in the experiment result of Fig. 18(b) can be captured by the TM-DH-BPD model. Additionally, there will be no interaction of the transverse and interface cracks when the coating thickness is higher ($h_{TC} = 300\ \mu$m, 500 μm). It can be concluded that whether or not the transverse and interface cracks interact is also determined by the coating thickness.

3.4.4. *The fracture behavior for different thermal shock surface*

At last, we will discuss the fracture behavior of TBCs for different thermal shock surfaces using the 3D model. The weak interface is assumed ($G_c^{TC/BC} = 0.001$ N/mm) and the coating thicknesses is fixed as $h_{TC} = 100\ \mu$m in this section. The damage patterns for different thermal shock surfaces and widths of TBCs during quenching are compared in Fig. 23. Three cases of thermal shock surfaces are considered. Case 1: The thermal shock on the upper surface of TBCs in Fig. 23(a, d); Case 2: The thermal shock on the upper, left and right surfaces of TBCs in Fig. 23 (b, e); Case 3: The thermal shock on the upper, left, right, front and back surfaces of TBCs in Fig. 23(c, f). When the width of TBCs is $w = 50\ \mu$m, the number of transverse cracks for the case 1 in Fig. 23 (a) is obviously more than that for the case 2 in Fig. 23 (b). It shows that the inclusion of thermal shock on left and right surfaces significantly influences the damage mechanism of TBCs when the width of TBCs is relatively small ($w = 50\ \mu$m). When the width of TBCs is relatively large ($w = 100\ \mu$m), the damage patterns in Fig. 23(d, e) for the two cases are similar.

Additionally, when the thermal shocks on the front and back surfaces are considered, things will be essentially different. To better show the interface crack patterns, TBCs are cut into a few slices in Fig. 23(c, f). Different from the first two cases that the TC/BC interface cracks initiate on the surfaces of the left and right ends and propagate along the x direction in Fig. 23(a, b, d, e), the TC/BC interface cracks for the case 3 initiate on the surfaces of the front and back ends and propagate along the z direction in Fig. 23(c, f). Since the length of TBCs in Fig. 19 is much larger than its width, the spalling failure of the TC/BC interface for the case 3 is more likely to occur before the appearance of the transverse cracks. It explains the important phenomenon that the interface cracking always occurs for different coating thickness and the transverse cracks may not exist during the thermal shock in the experiment of Fig. 18(a-c). When the thermal shock is applied on the surfaces of the front and back ends in the experiment, since the width of TBCs is relatively small in Fig. 23(c, f), the TC/BC interface crack will propagate along the direction of its width and completely cause the spalling failure of TBCs in a very short time. In this case, the damage patterns in Fig. 23 (c, f) are not influenced by the width of TBCs when the thermal shocks on the front and back surfaces of TBCs are considered.

Moreover, the transient temperature distributions and damage evolutions of TBCs for the width of $w = 100\ \mu$m are given in Fig. 24. With the transferring of the cold temperature from the surfaces of the front and back ends to the interior of TBCs, the TC/BC interface is immediately damaged at the time of $t = 0.4875$ ms. Meanwhile, the post processing video about the thermal and damage evolutions of TBCs under thermal shock for the 3D model can be seen in Appendix B (Video B.4). The total dual-horizon peridynamic element is 400,000. The computational time per analysis increment step and total memory usage for the 3D TBC model are 12.72 s and 9.6 gigabytes, respectively. Therefore, if the front and back ends of TBCs are suffered from the thermal shocks in the experiment, the 3D peridynamics-based thermo-mechanical damage model is needed to reasonably investigate the thermally-induced damage and interface debonding mechanisms of TBCs under thermal shocks. That is the reason why the authors establish the three-dimensional TM-DH-BPD model and also implement it in Abaqus. It may accelerate the

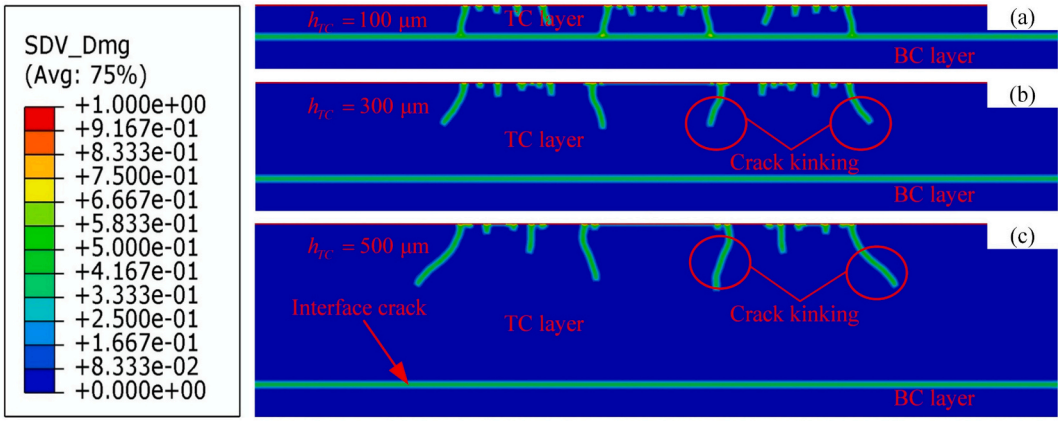


Fig. 22. The final damage patterns of TBCs for different coating thicknesses during quenching.

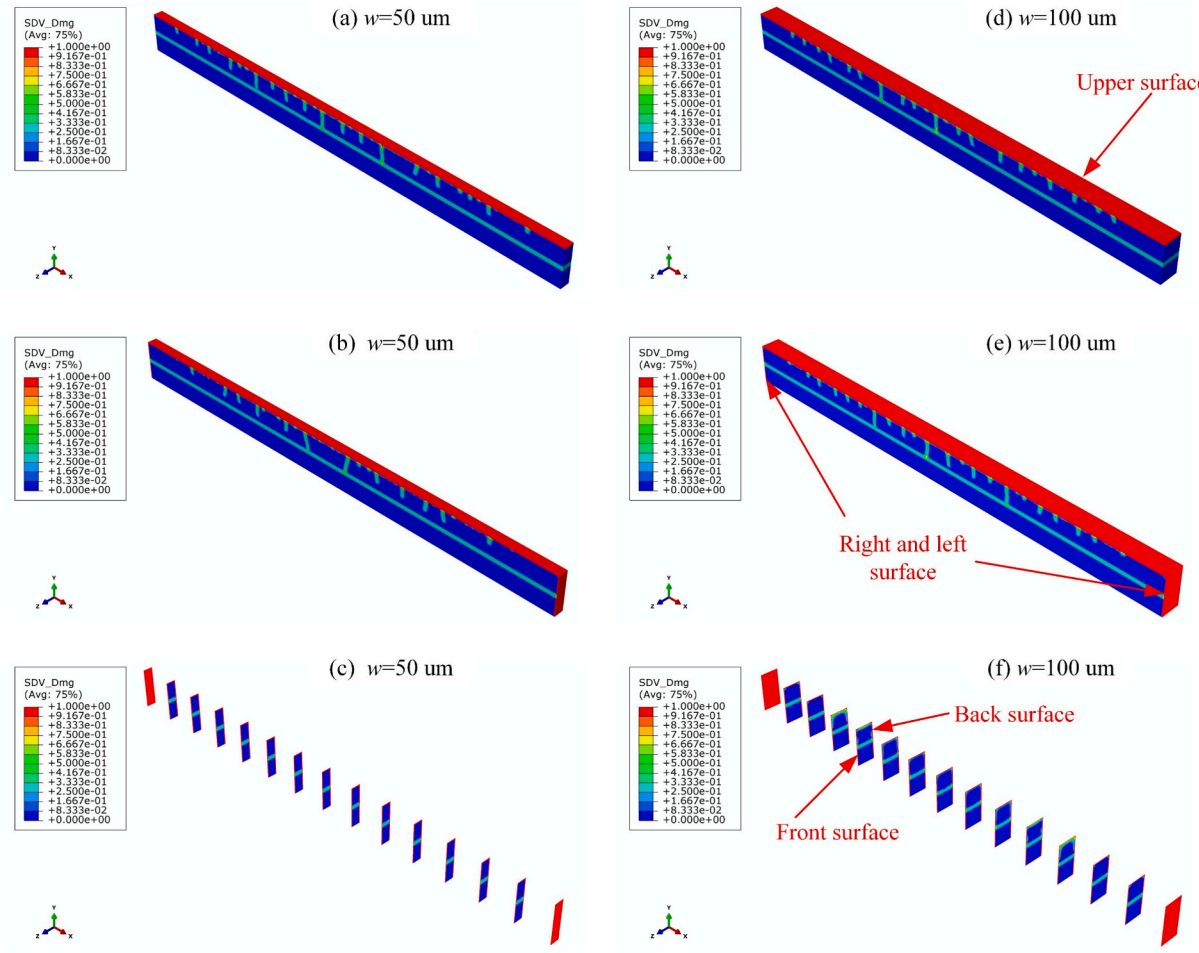


Fig. 23. The damage patterns for different thermal shock surfaces and widths of TBCs during quenching.

pace of dual-horizon peridynamics to deal with the real-world engineering problems including the complex interface debonding of TBCs under thermal shocks.

4. Conclusions

The coupled thermo-mechanical dual-horizon bond-based peridynamics (TM-DH-BPD) is proposed to address the thermal-induced fractures in thermal barrier coatings. Firstly, we investigate the effect of the mathematical forms of the micro-thermal conductivity for the TM-DH-BPD model on its numerical accuracy. Meanwhile, we give its optimum mathematical expression form by comparing with the numerical predictions of FEM. Additionally, we propose the dual-horizon peridynamic finite element method (DHPD-FEM) along with two approaches of the implicit solver and explicit solver with mass scaling to address the time scale gap. Moreover, the TM-DH-BPD model (including 2D and 3D models) has been firstly implemented in Abaqus UEL and VUEL subroutines. The dual-horizon peridynamics-based coupled thermo-mechanical elemental stiffness matrices and residuals in Abaqus/UEL and the coupled thermo-mechanical elemental internal forces and mass matrices in Abaqus/VUEL are provided in detail for the researchers and engineers. Therefore, the built-in powerful solver and post processing capabilities of Abaqus can be used. At last, we use the TM-DH-BPD model to investigate the thermally-induced fracture mechanism of TBCs, the phenomena of the transverse cracking propagation, kinking and its interaction with the TC/BC interfacial cracking observed in the experiment are well captured. To provide more detailed information about the crack propagation and interaction, some corresponding post processing videos are given. The proposed TM-DH-BPD model and its finite element implementation in Abaqus may accelerate the pace of the theoretical researching and real-word application of peridynamics including the complex thermal shock cracking problems of TBCs.

The proposed TM-DH-BPD model is of effectiveness to preliminarily reveal the key thermally-induced fracture phenomena observed in the experiment of TBCs. There is still need for further development based on the present study in the future:

(1) High temperatures can significantly reduce the mechanical strength and thermal stability of TBC materials (e.g., oxidation of the bond coat or sintering of the top coat). Additionally, creep deformation of TBCs under long-term high temperatures can relax thermal stresses, alter crack driving forces, and thus influence crack propagation paths and rates. It can be seen that the deformation and damage of TBCs are very complex. The further development of the TM-DH-BPD model is needed to consider the temperature-dependent materials, oxidation of the bond coat, creep effects and sintering of the top coat. It will be extended to study the thermally-induced damage propagation and interface debonding under temperature cycling in the future.
(2) The interfacial debonding under thermal shock typically exhibits a mixed-mode failure pattern. To accurately predict realistic interfacial failure behaviors, it is very valuable to establish more refined interface fracture criteria with the help of experiment to distinguish the physical mechanisms of Mode I and Mode II fractures at interfaces.

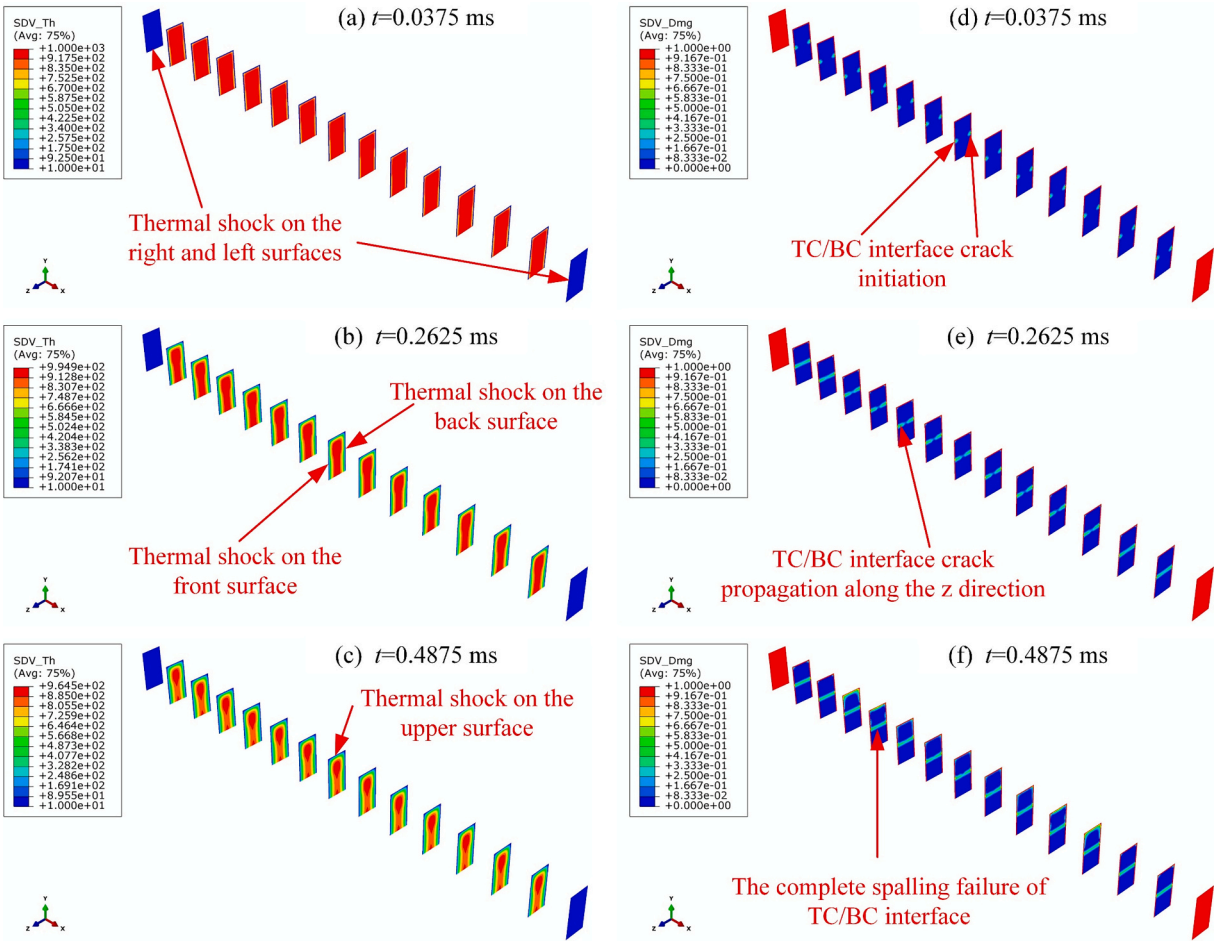


Fig. 24. The thermal and damage evolutions of TBCs during quenching.

(3) Using the same framework of DHPD-FEM and the definition of dual-horizon peridynamic element, it is convenient to implement other peridynamic formulation (e.g., non-ordinary state-based peridynamics, PDDO [79,80] et al.) or NOM [81–83] (that allows more general constitutive models) into Abaqus. Which are the directions for the further work to capture more complex real-world phenomena, such as anisotropic material behavior or multi-layered structures.

(4) Using the similar method of implementing the dual-horizon peridynamics-based coupled thermo-mechanical model in Abaqus UEL and VUEL subroutines, other similar multi-physical coupled models (e.g., mechano-chemical, thermo-mechano-chemical and thermo-electrical et al.) can be solved by the proposed the dual-horizon peridynamic finite element method. These coupled models can be easily implemented in Abaqus to address the corresponding complex coupled multi-physical engineering problems with the interface damage propagation in the context of peridynamics.

The authors believe that the dual-horizon peridynamic finite element implementation of coupled multi-physical damage models in Abaqus provides a new and effective theoretical and computational method in applied fracture mechanics and engineering for researchers and engineers. It may accelerate the pace of the theoretical researching and real-word application of peridynamics.

CRediT authorship contribution statement

Yehui Bie: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huilong Ren:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Data curation, Conceptualization. **Ya Duan:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Tinh Quoc Bui:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Xiaoying Zhuang:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Erdogan Madenci:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Timon Rabczuk:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Yueguang Wei:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Deriving process of the micro-thermal conductivity function

The peridynamics-based pairwise heat flow density function f_θ [70,71,84] is written as

$$f_\theta = \frac{\partial \phi(\mathbf{x},\mathbf{x}')}{\partial \underline{\theta}\langle\boldsymbol{\xi}\rangle} = k(\mathbf{x},\mathbf{x}')\cdot\frac{\theta(\mathbf{x}')-\theta(\mathbf{x})}{|\mathbf{x}'-\mathbf{x}|^\beta} = \frac{k(\mathbf{x},\mathbf{x}')\cdot\underline{\theta}\langle\boldsymbol{\xi}\rangle}{|\boldsymbol{\xi}|^\beta} \tag{A.1}$$

where $\phi(\mathbf{x},\mathbf{x}')$ denotes the micro-thermal potential between two material points of $\mathbf{x}$ and $\mathbf{x}'$, which is the function of the temperature state $\underline{\theta}\langle\boldsymbol{\xi}\rangle = \theta(\mathbf{x}')-\theta(\mathbf{x})$. $k(\mathbf{x},\mathbf{x}')$ denotes the micro-thermal conductivity. The shape parameter of the micro-thermal potential β is constant. When $\beta = 1$, the pairwise heat flow density function f_θ is consistent with that defined by Oterkus et al. [84]. When $\beta = 2$, it is consistent with that defined by Bobaru et al. [70,71].

From Eq. (A.1), the micro-thermal potential $\phi(\mathbf{x},\mathbf{x}')$ can be expressed as

$$\phi(\mathbf{x},\mathbf{x}') = \frac{k(\mathbf{x},\mathbf{x}')\cdot\left(\underline{\theta}\langle\boldsymbol{\xi}\rangle\right)^2}{2\cdot|\boldsymbol{\xi}|^\beta} = \frac{k(\mathbf{x},\mathbf{x}')\cdot(\theta(\mathbf{x}')-\theta(\mathbf{x}))^2}{2\cdot|\boldsymbol{\xi}|^\beta} \tag{A.2}$$

The nonlocal peridynamics-based thermal potential $\overline{\Phi}$ is calculated as

$$\overline{\Phi} = \frac{1}{2}\int_{\mathscr{H}_x}\phi(\mathbf{x},\mathbf{x}')\mathrm{d}V_{x'} \tag{A.3}$$

For 2D case, the linear temperature distribution $\theta(x,y) = x+y$ is assumed. Substituting $\theta(x,y) = x+y$ into Eqs. (A.2) and (A.3),

$$\overline{\Phi} = 1\Big/2\cdot H\cdot\int_0^{2\pi}\int_0^{\delta_x}\left(1\Big/2\cdot k\cdot(\xi\cdot\cos\theta+\xi\cdot\sin\theta)^2\cdot\xi\right)\Big/\xi^\beta\mathrm{d}\xi\mathrm{d}\theta;\quad \xi = |\boldsymbol{\xi}| \tag{A.4}$$

where H denotes thickness for the 2D cases.

The local thermal potential Φ is defined as

$$\Phi = \frac{1}{2}\kappa\cdot\nabla\theta\cdot\nabla\theta \tag{A.5}$$

where κ denotes the thermal conductivity coefficient.

Substituting $\theta(x,y) = x+y$ into Eq. (A.5), the local thermal potential is $\Phi(\mathbf{x}) = \kappa$. Since the nonlocal peridynamics-based thermal potential $\overline{\Phi}$ and local thermal potential Φ is equivalent, the micro-thermal conductivity can be solved by combining Eqs. (A.4) and (A.5). When the constant micro-thermal conductivity is assumed, the micro-thermal conductivity $k(\mathbf{x},\mathbf{x}')$ is derived as

$$k(x,x') = \begin{cases} 4\kappa\big/\left(\pi\cdot H\cdot\delta_x^2\right);\beta = 2 \\ 6\kappa\big/\left(\pi\cdot H\cdot\delta_x^3\right);\beta = 1 \\ 7\kappa\big/\left(\pi\cdot H\cdot\delta_x^{3.5}\right);\beta = 0.5 \end{cases} \tag{A.6}$$

Bobaru et al. [70,71] states that the shape parameter γ will influence the numerical accuracy of the peridynamics-based heat conduction model. When the nonlinear micro-thermal conductivity is assumed, $k(\mathbf{x},\mathbf{x}') = k_0(1-(\xi/\delta_x)^\gamma)$; it can be derived for the shape parameter of $\gamma = 1$:

$$k(x,x') = \begin{cases} 12\kappa\big/\left(\pi\cdot H\cdot\delta_x^2\right)\cdot(1-\xi/\delta_x);\beta = 2,\gamma = 1 \\ 24\kappa\big/\left(\pi\cdot H\cdot\delta_x^3\right)\cdot(1-\xi/\delta_x);\beta = 1,\gamma = 1 \\ 63\kappa\big/\left(2\cdot\pi\cdot H\cdot\delta_x^{3.5}\right)\cdot(1-\xi/\delta_x);\beta = 0.5,\gamma = 1 \end{cases} \tag{A.7}$$

Similarly, it can be derived for the shape parameter of $\gamma = 2$:

$$k(x,x') = \begin{cases} 8\kappa\Big/\left(\pi\cdot H\cdot\delta_x^2\right)\cdot\left(1-(\xi/\delta_x)^2\right);\beta = 2,\gamma = 2 \\ 15\kappa\Big/\left(\pi\cdot H\cdot\delta_x^3\right)\cdot\left(1-(\xi/\delta_x)^2\right);\beta = 1,\gamma = 2 \\ 77\kappa\Big/\left(4\cdot\pi\cdot H\cdot\delta_x^{3.5}\right)\cdot\left(1-(\xi/\delta_x)^2\right);\beta = 0.5,\gamma = 2 \end{cases} \tag{A.8}$$

For the 3D case, the linear temperature distribution $\theta(x,y,z) = x+y+z$ is assumed. Substituting $\theta(x,y,z) = x+y+z$ into Eqs. (A.2) and (A.3),

$$\overline{\Phi}(\mathbf{x}) = 1\Big/2\cdot\int_0^{\pi}\int_0^{2\pi}\int_0^{\delta_x}\left(1\Big/2\cdot k\cdot\left(f_\xi\right)^2\cdot\xi^2\cdot\sin\varphi\right)\Big/\xi^\beta\mathrm{d}\xi\mathrm{d}\theta\mathrm{d}\varphi \tag{A.9}$$

$$f_\xi = \xi\cdot\sin\varphi\cdot\cos\theta+\xi\cdot\sin\varphi\cdot\sin\theta+\xi\cdot\cos\varphi;\xi = |\boldsymbol{\xi}| \tag{A.10}$$

Similarly, substituting the linear temperature distribution $\theta(x,y,z) = x+y+z$ into Eq. (A.5), the local thermal potential is $\Phi(\mathbf{x}) = 3\kappa/2$. The micro-thermal conductivity $k(\mathbf{x},\mathbf{x}')$ can be derived by equating the local and nonlocal thermal potentials for different shape parameters of γ and β. Take $\gamma = 0.5$, $\beta = 1$ as an example, the mathematical expression of the micro-thermal conductivity $k(\mathbf{x},\mathbf{x}')$ for 3D case can be obtained as

$$k(\mathbf{x},\mathbf{x}') = 54\kappa \Big/ \left(\pi \cdot \delta_x^4\right) \cdot \left(1 - \left(\xi/\delta_x\right)^{0.5}\right);\ \ \beta = 1, \gamma = 0.5 \tag{A.11}$$

Appendix B. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tafmec.2025.105298.

Data availability

No data was used for the research described in the article.

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